

# **Effect of Social Audits on MGNREGA Program Delivery in Andhra Pradesh**

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by Amol Shaila Suresh

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## Declaration

I, the undersigned, hereby affirm and declare that the dissertation titled “Effect of Social Audits on MGNREGA Program Delivery in Andhra Pradesh” has been written by me under the direct guidance of my supervisors, Dr. Arup Mitra and Dr. Namrata Gulati, and has been submitted in partial fulfilment of the requirements of M.A (Development Economics) Program of South Asian University. I further declare that the research work is original and I assume responsibility for the results and findings in the paper. This work has not been previously submitted for any other degree program.

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# **Effect of Social Audits on MGNREGA Program Delivery in Andhra Pradesh**

## **Abstract**

Using panel data of five years extracted from official reports of SSAAT and MIS, we study the impact of social audits on MGNREGA program performance in Andhra Pradesh. The data for over 650 mandals of 13 districts was considered for the analysis. We find that social audit frequency has an insignificant effect on performance parameters of the MGNREGA, but other variables associated with social audit have significant positive effect on the program performance. The program performance has seen the most significant positive effect in covid years, due to high demand for work and increased budget. We also find that social audit process exposes financial mismanagement and corruption in the MGNREGA. This highlights that social audits need to be promoted in other public schemes as a tool to achieve accountability from public officials. Our results are suggestive of positive influence of social audits on program performance, but they are not conclusive.

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Any errors, omissions, or shortcomings in this thesis are solely my responsibility.

**Amol Shaila Suresh**

# CHAPTER 1

## Introduction

All developing countries face the challenge of translating a good-intended public scheme/ program into intended outcomes and benefits. There are many issues like leakages, corruption, lack of competency and limited capacity of government officials, in developing countries. An Indian economist famously said in a column that it is better to drop money from the helicopter or gas balloon over rural areas, than formulating a public scheme for poverty reduction (Aiyar, 2004 December 18). There is also a school of thought that believes a decentralized governance structure with local participation and voice from the grassroots can resolve the issues pertaining to the programme quality. This dissertation looks into the effect of social audit, a participatory monitoring approach, on the quality of MGNREGA – a flagship employment programme of Government of India.

### 1.1 Mahatma Gandhi National Rural Employment Guarantee Scheme

India, since its Independence in 1947, has seen many poverty alleviation programmes. Among them, employment generation schemes have been integral to the poverty reduction effort. The state of Maharashtra was the pioneer to start an employment guarantee program for rural population during desperate times of early 1970s drought (Kajale & Shroff, 2018). Following which many programs like Food for Work Program (1977), Rural Landless Employment Guarantee Programme (1983), Jawahar Rozgar Yojana (1989) and Employment Assurance Scheme (1993). These schemes were modified given the emphasis on rural infrastructure development started growing. It culminated into Jawahar Gram Samridhi Yojana

(1999), Sampoorna Grameen Rozgar Yojana (2001) and finally National Food for Work Programme (2004).

On 7 September 2005, the Government of India passed the Mahatma Gandhi National Rural Employment Guarantee Act, which gives any adult member of a rural household legal guarantee of a hundred days of employment. The employment provided in MGNREGA includes unskilled manual labour like creation of common village infrastructure and assets, maintenance of public goods like ponds, canals, roads, watershed work, forest, etc. The major difference between the employment programmes enlisted earlier and MGNREGS is that latter sees employment as a right and the Government of India is legally binding to provide it to whoever makes the demand, within 15 days. In case government fails to provide the work to the individual within 15 days, then individual is entitled to an unemployment allowance.

Mahatma Gandhi National Rural Employment Guarantee Scheme is the largest social safety nets programme in the world (World Bank, 2015). Its design is such that MGNREGA acts an insurance/ safety net for households in lean economic period or economic crisis/ shock. For example, in Covid-19 pandemic, the MGNREGS provided protection to many vulnerable households. In a recent survey conducted by Centre for Sustainable Employment, Azim Premji University and NREGA Consortium (2022), it is quite evident that MGNREGS protected many extremely poor families from the economic shock. Almost 80-90% of the surveyed families reported an income loss, in the range of 30-50%. While wage payment through MGNREGA compensated for 20-80% income loss during Covid-19 pandemic (Azim Premji University, 2022). As, MGNREGA targets and provides

employment to the most vulnerable section, rural unskilled labour, the programme is one of the most significant among many poverty reduction programmes.

## **1.2 Participatory Approach in Development**

There are more than 6 lakh villages in India and 69% of the population lives in rural India (2011 Census of India). Government of India has taken a few strides in making governance structure more inclusive to the population living in rural areas, far away from the power centres, through the 73rd and 74th Constitutional Amendment in 1992. Decentralization of political power to Panchayat Raj institutions was a major step in the direction of making ordinary people participate in politics and governance of their villages and locality. Until 73rd Amendment, village institutions were hollow & ineffective and monopolized by local elite; but after the decentralization measures taken by the Union and State governments, new institutions have been established, new decision-making procedures have been developed, and new channels of participation have been made available (Heller, Harilal & Chaudhari, 2007). Similarly, the Forest Rights Act (2006) or Panchayat Extension to Scheduled Areas Act (1996) endow tribal communities rights to participate in the decision making of the forest ecosystem. South Asian region has many important and innovative community-based forestry initiatives which explain why and how poorest of the poor people, given a say in the governance, can establish efficient management systems over millions of hectares of forestlands, facilitating rapid regeneration (Poffenberger, 2000). However, mere devolution of power to local communities cannot ensure intended participatory governance. The local elites and bureaucracy will retain their control and remain de facto administrator, citing reasons like local incompetence (Ribot, 2009). When we use the word "participatory," we refer to various forms and levels of participation,

control, and decision-making in an activity or a process. The rationale of participatory approach is to enhance the effectiveness and transparency in the process.

### **1.3 Social Audit in MGNREGA**

There is a history of government-led audits of government programmes and schemes. India has enacted Comptroller and Auditor General of India Act, 1971 as an independent auditing entity to review the functioning of various ministries and departments of Government of India and state governments. However, these measures are centralized and leave out ordinary layperson from actively participating in the process of evaluation as it's conducted by government officials or technical experts. On the other hand, India has also seen many initiatives regarding an inclusive approach to governance where local beneficiaries actively take part in planning, monitoring and evaluation of the programme.

Social Audit is a such approach in the sphere of public policy and governance. Social audits have been considered as an effective means of improving implementation of many social protection programs like Integrated Child Development Scheme (ICDS), Public Distribution System (PDS), Mid-day Meal Scheme, etc. Social audit was firstly developed by civil society organizations like Mazdoor Kisan Shakti Sangathan (MKSS) under the right to information movement. Social audit is a process of monitoring of a programme that is conducted by individuals, particularly those who are impacted or are intended beneficiaries of the programme being audited, and facilitated by government (MoRD, 2016 19 December). MGNREG Act mandates conducting social audit as a post-implementation exercise at least once every six months to monitor all activities under MGNREGS. Rule 4 of Audit of Schemes Rules, 2011 specifies that each State

Government shall identify or establish Social Audit Unit (SAU), an independent organization, for facilitating social audit of MGNREGS works (MoRD, 2013). An elaborate process of social audit includes verification of all document proofs like muster rolls, registers, material bills, inspection of status and quality of works, investigating discrepancies and complaints raised by the workers, convening a public meeting and reading out details of all projects, discussing the finding in the public gathering. A special *Gram Sabha* is called for conducting MGNREGA Social Audit where members of *Gram Sabha* can express their grievances. State governments are responsible for setting up Social Audit Unit (SAU) which will give training to the resource persons for conducting social audit process.

An effective social audit process can give boost to grassroot democracy and community activism at the village level, and create a demand for collective accountability. The participatory approach of social audit will also improve programme implementation quality, by controlling the leakages and exercising entitled rights within the law. In this light, this dissertation will try to look at the effect of social audit process on the programme implementation quality of MGNREGS – trends of wage and work-related complaints, completion of 100 days of work, etc. Additionally, this dissertation will also try to discuss the effect of social audit on the strengthening of local democracy and exercising own rights. In the next section, Section II, a detailed literature review of studies and experiments conducted on social audit process of various social schemes is given. Section III states the research question that this dissertation will try to address, which will be followed by a discussion on dataset and methodology in Section IV.

## **CHAPTER 2:**

### **Analytical Framework**

The Mahatma Gandhi National Rural Employment Guarantee Act is a pivotal social welfare program designed to provide employment to rural households. However, there have been concerns about the effectiveness of the program in providing sustainable livelihoods to rural families. Social audit is one of the mechanisms introduced to enhance transparency and accountability in implementing MGNREGA. A social audit is a participatory approach to monitoring and evaluation that involves the active engagement of local communities in assessing the performance of public programs. This thesis proposes an analytical framework to understand the impact of social audits on public welfare programmes, like MGNREGA.

The proposed analytical framework seeks to examine the impact of social audits on various dimensions of MGNREGA performance due to transparency, accountability and public participation. While MGNREGA gives citizens the right to demand work from the state, the social audit process enables citizens to become more engaged in the implementation and monitoring of the program. It is expected that social audits can improve program implementation and reduce corruption by increasing the transparency and accountability of MGNREGA. The theoretical framework, to look at this effect of social audit on program performance, is based on the accountability theory. Accountability theory is based on the principle that public officials and institutions are responsible for fulfilling their duties and obligations to the public, and that they are held accountable for their actions and decisions. This theory recognizes the importance of transparency, participation, and

feedback in promoting good governance, and emphasizes the need for institutions to be responsive and adaptive to changing social, economic, and political conditions (Bovens, 2005; Dubnick, 2005; Mulgan, 2000).

The key elements of accountability theory include transparency, participation, and feedback. Transparency refers to the availability and accessibility of information about government policies, decisions, and actions. Participation refers to the involvement of citizens and stakeholders in the design, implementation, and evaluation of government programs and services. Social audits can increase the participation of vulnerable and marginalized groups in MGNREGA by raising awareness of their entitlements under the act. Feedback refers to the use of information and data to monitor, evaluate, and improve government performance (Bovens, 2005; Dubnick, 2005; Mulgan, 2000).

Social audits are an important mechanism for promoting accountability in government programs and services. Social audits involve the independent review and assessment of program performance by citizens and stakeholders, and can provide valuable information and feedback to improve program design and implementation. Social audits also promote transparency by making information about program performance available to the public, and can enhance citizen participation by providing opportunities for feedback and input (Chhotray & Stoker, 2010; Fox & Brown, 2014; Joshi, 2014).

The theoretical framework of accountability theory provides a useful lens through which to examine the impact of social audits on the MGNREGA scheme's performance and leakages in Andhra Pradesh. The study will explore the extent to which social audits promote transparency, participation, and feedback in the implementation of the MGNREGA scheme, and will examine the relationship

between social audits and various indicators of program performance, such as wage expenditure, average person days of work provided, households completing 100 days of work, job cards generated, delayed wage payment, financial deviation, recovery, penalty imposed, and issues reported and resolved during social audits.

The social audit process rejuvenates the Panchayat Raj system, where local communities participate in the monitoring of rural development programs such as MGNREGA. Thus, social audits of public welfare schemes can potentially transform local communities into empowered actors who can hold the state accountable for delivering public goods and services effectively. Table 1 illustrates how social audits can act as an empowering mechanism for local communities, by providing them with a voice and platform to demand greater transparency and accountability in the implementation of MGNREGA. The table proposes four key factors of empowerment - inclusion, information, influence, and integration (Vij, 2012). Inclusion refers to the participation of members of all strata of society, particularly marginalized communities. Forming a representative social audit committee is a crucial step in ensuring that all voices are heard. The information factor entails educating communities about their rights and entitlements under MGNREGA, especially to the marginalized sections discriminated against based on lack of information.

**Table 2.1: Social Audits as an Empowerment Process**

| <b>Stages of empowerment process</b> | <b>Social Audit Process</b>                                                                       |
|--------------------------------------|---------------------------------------------------------------------------------------------------|
| Inclusion                            | <ul style="list-style-type: none"> <li>• Forming representative Social Audit Committee</li> </ul> |
| Information                          | <ul style="list-style-type: none"> <li>• Proactive disclosure of information online</li> </ul>    |

|             |                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <ul style="list-style-type: none"> <li>• Documents submitted before Social Audit Committee at least 15 days before the audit</li> <li>• Information can also be sought through Right to Information Act</li> <li>• Resource groups to provide training to SAC to conduct social audit</li> </ul>                                                                                                 |
| Influence   | <ul style="list-style-type: none"> <li>• Social audit meeting is open to public participation</li> <li>• Gram Sabha members can ask questions and seek redressal</li> <li>• Media, civil society organizations participate as external observers</li> <li>• If discrepancies or grievances are legitimate, implementing agency must commit to take appropriate action within timeline</li> </ul> |
| Integration | <ul style="list-style-type: none"> <li>• An action taken report to be submitted</li> <li>• Social audit committee can approach Grievance Redressal or Ombudsman Cell and high administration</li> </ul>                                                                                                                                                                                          |

*Source: Vij, N. (2012)*

The influence factor involves empowering communities to hold government officials accountable for their actions and ensuring that grievances are resolved in a timely manner. The members of gramsabha can pose questions to government officials during social audits, and hold them accountable for any discrepancies or corruption in the implementation of MGNREGA. The presence of external independent observers can further ensure the fairness and objectivity of the social audit process. The discrepancies and grievances highlighted during social audits are documented and passed on to implementing authorities for further action, providing a basis for redressal and corrective measures. The integration factor refers to the integration of social audits into the larger governance framework of MGNREGA, where social audits become a key tool for monitoring and evaluating program implementation at all levels. The implementing agencies need to submit an Action Taken Report

on the social audit findings within a month, otherwise social audit committee can escalate the matter to Grievance Redressal or Ombudsman Cell and higher administration. The proposed framework for social audits under MGNREGA aims to facilitate the empowerment of local communities by providing them the opportunity to participate in decision-making and demand greater transparency and accountability in the implementation of MGNREGA, through inclusive representation, education on rights and entitlements, holding officials accountable, documenting discrepancies and grievances, and integrating social audits into the governance framework.

## **CHAPTER 3**

### **Literature Review**

In this section, we will first discuss about the performance of MGNREGS, along the lines of procedural flaws, outcomes and leakages. In the next section, we will discuss participatory monitoring and evaluation in general, the objective of conducting PM&E. This is followed by a discussion on social audit, a specific example of participatory M&E, practised for various public welfare schemes and programmes in different parts of the world and in India. Through this section, we try to understand the existing literature on social audit process and impact of social audit on NREGS performance.

#### **3.1 Performance of MGNREGS**

The largest employment guarantee scheme in the world, Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS), has given millions of unskilled poor Indians crucial financial support since its inception. But the legal mandate of providing 100 person-days of work per household still remains unfulfilled. In practice, the provision of work is based on supply, rather than demand for work, which is against the spirit of ‘right to work’ underlying MGNREGS (Dutta et al., 2014; Desai, 2015). In the field survey conducted in four states of India during pandemic induced lockdown, respondents reported that they would have liked to work for 64 more days on average (Azim Premji University [APU], 2022). Desai (2015) also suggested that there is further rationing in the amount of work who manage to get some work. Dutta et al. (2014) argue that unmet demand is the single most critical factor in addressing the gap between actual scheme performance

and scheme's intended impact on poverty. APU (2022) reported that roughly 39 percent of all the eligible job card holding households, interested in working under NREGS, could not get a single day of work. Further, it is revealed by 63 percent respondents that sufficient work not being sanctioned or opening up is the biggest reason for the unmet demand (APU, 2022). There are frequent complaints of delay in payments of wages, which is a significant reason of decline in demand for MGNREGA work (Ehmke, 2015). Chopra (2015) discussed that the lack of administrative and technical capacity for measuring the work done at each work site, calculating wages leads to delays and decline in payments. Aiyar and Mehta (2013) have provided a detailed account of corruption happening in MGNREGS. A nexus between local elites and MGNREGS staff may make up an entire work site and appropriate most of the budget among them (Aiyar & Mehta, 2013). Dutta et al. (2014) reports that most of the beneficiaries are not aware of the process of obtaining work under the scheme. Many studies of inter-state comparison of MGNREGA performance reveal that Rajasthan, Madhya Pradesh, Andhra Pradesh record the highest percentage of households completing 100 days of work (Ahmed, 2021). There is a lot of variation in the performance of MGNREGA implementation and states can be ranked according to their performance (Turangi, 2022). This dissertation will try to understand if frequent and better social audit implementation have any effect on the state's MGNREGA performance.

### **3.2 Participatory Monitoring and Evaluation**

In contrast to traditional M&E, participatory monitoring and evaluation aims to involve all stakeholders in every step of the process. The participatory monitoring and evaluation (PM&E) has developed over a period of last 20 years of using participatory research in development. PM&E draws from the traditions of various

participatory approaches such as Participatory Action Research (PAR) led by Paulo Freire in 1972, Participatory Rural Appraisal (PRA) by Robert Chambers in 1997 and few others. The main objective of participatory M&E is to provide stakeholders of the program with complete knowledge of whether the project goals have been achieved and how resources have been used, for improving program implementation and decisions relating to project finance (Campos and Coupal, 1996; PRIA, 1995). Participatory M&E emphasises a process of sequential reflection and action, carried with and by local people. Local knowledge and viewpoints are not only taken into consideration but also serve as the foundation for planning and research (Cornwall and Jewkes, 1995). Holte-McKenzie, Forde & Theobald (2006) talked about why it is important to respectfully engage with communities & other stakeholders and importance of engagement youth in the participatory M&E in a Kenya-based study. A community-based monitoring and evaluation increases public accountability by building capacity of local people and imbibing research and monitoring skills into them such that they will hold high-level authorities accountable (Parachini and Mott, 1997). Bradley et. al. (2002) discusses participatory strategies used by international health organization to collaborate with local community for improving quality of reproductive health, identifying issues in the health facilities, and defining quality care from their perspective. Problems of local stakeholders can be solved in a sustainable and continuous quality improvement manner through long-term participatory techniques. A randomized field experiment in Uganda suggests that community-based monitoring of public health care providers improved health outcomes – reduced child mortality and increased child weight (Björkman and Svensson, 2009). The participatory M&E can add value to the research by building trust among researchers, local community and government officials, increasing opportunities of democratic participation for oppressed groups like women &

tribals, building a sense of ownership and stronger community, formulating clearer goals considering diversity in the community, and increasing space for reflection and adaptation (Vernooy, Qiu, & Xu, 2006).

### **3.3 Social Audit**

Social audit has become a tool of making the government accountable and transparent through community participation, and it is a significant medium strengthening decentralization and deepening local democracy. Social audit, in principle, serves a purpose of democracy like informing citizens, encouraging their participation and creating a sense of responsibility to collectivize & address issues (Chandoke, 2007). Social audit is intended to empower people to use the information collected from the Right to Information Act to question the government officials and create pressure for them to respond to the grievances and charges of wrongdoings (Aiyar, 2010). Just not merely focusing on the accounts and reports, social audit emphasizes on open & transparent dialogue, agenda setting and feedback, while people being facilitator and moderator, unlike an independent auditor (Cotton, Fraser & Hill, 2000).

Swain and Sen (2009) have documented the experiences of using social audits for Integrated Child Development Services (ICDS) Scheme in Andhra Pradesh and Orissa. The implementation of social audit for ICDS has resulted into reduction of stockout period of nutritional supplement from average 20 days to zero. Social audit ensured that ASHA, who are better trained than AWW, were assigned as the first point of contact for pregnant women (Swain & Sen, 2009). Sinha (2008) examined the use of social audit for Mid-day Meal Scheme by the government of Andhra Pradesh. She found out that social audit brought out many issues in public like corruption, fund mismanagement and misuse of power. Social audits for MDM in

Andhra Pradesh ensured a cooked midday meal served every day in all schools; eggs were served for the first time. Social audits facilitated mobilization of villagers for children's rights around MDM, increased attendance in *Gram Sabha* and immediate decision making for children's midday meal (Sinha, 2008). A study conducted in Dang district of Nepal showed that social audit process increased timely opening and closing of health facilities, quality & accessibility of services and financial transparency (Gurung et al, 2019). The irregularities in the expenditure of completed works of NREGS has down from 1.74 percent to 0.40 percent after three rounds of social audit (Tambe et al., 2016). In a study done in South Africa by Adota et al. (2005) argued that the community participation in government programmes has a strong impact on the scheme performance, measured in terms of log number of days of work created, log number of days of training, and share of the budget spent on labour. Afridi and Iversen (2014) have reported in their findings of Andhra Pradesh social audit model that there is a steep rise in complaints about delayed payment or non-payment of wages after three rounds of social audits. On the other hand, complaints regarding non-provision of work decline in the same period. The program expenditure is reported to be more than doubled just after the first social audit (Afridi & Iversen, 2014). While for employment generation, total number of person days steeply increased almost threefold between first and third social audit (Afridi & Iversen, 2014). Just the exposure to or participation in social audit process can likely augment the awareness about MGNREGA rights and entitlements under the scheme (Aiyar et al., 2013). The existing literature on social audits clearly suggests that the social audit has a potential to strengthen democratic voice at the grassroots and improve scheme performance.

Though social audit appears as a perfect design for detecting corruption, controlling misconduct, strengthening local democracy, and making responsible and responsive citizens, but just like any other programme, it doesn't always translate into an ideal picture. According to the Social Audit Status Report of the Ministry of Rural Development, 13 percent of districts didn't conduct any social audit in the year 2010-11. Even if they are conducted, it's highly likely that they took place only on record or conducted dishonestly. In practice, there is a large variation in number and quality of social audits across states. It is observed that most states have implemented social audits sporadically (CAG, 2013).

Cleary, Molyneux and Gilson (2013) argues that the lack of technical expertise, lack of clarity in roles and responsibilities and low education levels limits the effectiveness of accountability initiatives. Afridi (2008) in her observations from a study of MGNREGA social audits in Rajasthan and Andhra Pradesh reported that ordinary villagers are intimidated by local powerful elites; hence guidance and support of local NGOs becomes very significant while conducting social audits in the villages. Most men and women were not cognizant of the precise details of rights and entitlements under the scheme, even though most of them were aware about NREGA (Dutta et al., 2014). As beneficiary learning mainly takes place "by doing" and happens with a time-lag, it hinders the effectiveness of participatory audits (Afridi & Iversen, 2014). Like beneficiaries, transgressors also learn by doing new ways and avenues for wrongdoings and they always stay one step ahead of the auditors (Olken and Pande, 2011). In an EPW article, Gopal K S (2009) debates that despite legal mandate of the Right to Information Act, there has been persistent delays in provision of information to social audit bodies relating to scheme. The real-time information is not readily made available publicly for scrutiny. Aiyar, Mehta and Samji (2013) advocate for administrative reforms to

punish government officials who are involved in the misappropriation of funds, as accountability without backing of enforcement in case of social audits will have no use. Without substantive disciplinary action is taken against the officials found guilty of corruption or wrongdoing, the credibility, effectiveness and enthusiasm of the people in social audit will not sustain (Afridi, 2008). Though social audit model has been successful in detecting misdeeds and irregularities in funds, social audit fails to deter these wrongdoings from happening (Afridi and Iversen, 2014).

We can see there is mix evidence of effectiveness of social audit. It is clearly seen from the studies mentioned above that social audit is effective in detecting corrupt practices. The most of the existing literature on social audits in NREGS has primarily focussed on only state – Andhra Pradesh. This dissertation will try to look beyond corruption and focus on program implementation quality by looking at the data from various states.

### **3.4 Research Questions**

The literature review gives us a clear sense that social audit is an important exercise for reducing leakages in government programmes, increase people's participation in the governance and make government machinery accountable to people. There is a significant amount of work done on the effect of social audits on leakages in public schemes. A few studies have looked into the impact of social audits on programme quality like complaints regarding non-delay of payments, provision of work, etc. There are a few qualitative studies which have reported that participating in social audit has created awareness about rights and entitlements among the villagers. The literature doesn't much talk about impact of social audits on programme related outcomes, strengthening of democratic voices, and confidence in villagers to speak to authorities in power.

1. *Does carrying regular social audits of MGNREGS works mandated in the Act have any impact on the leakages/ corruption/ misappropriation of funds? Do the leakages reduce further with the number of social audits?*

Most of the public offices and schemes in developing world are infected by corruption or some sort of leakages. This question will help us understand whether public monitoring of MGNREGS by the direct beneficiary of the program will have any desirable effect on the leakages and punitive action taken against the transgressors. Afridi (2008) have studied social audit of Rajasthan and Andhra Pradesh to analyse the effect of social audit on leakages, however the work is limited to a few GPs of two districts only. This dissertation will try to analyse the data of districts and GPs from various other states.

2. *Does carrying regular social audits of MGNREGS works mandated in the Act have any impact on quality of programme implementation? Does the quality improve further with the number of social audits?*

This question will help us understand whether public vigilance and monitoring help improve quality of NREGS implementation. In the programme implementation quality, we will mainly focus on labour & material complaints. Afridi and Iversen (2014) have look into the social audit data of 300 GPs of Andhra Pradesh to assess the effect of multiple social audits on program related complaints. This dissertation will use the similar approach to analyse the data of more diverse set of GPs and states.

3. *Does carrying regular social audits of MGNREGS works mandated in the Act have any impact on programme outcomes? Do programme outcomes improve with the number of social audits?*

Very limited research has been done on analysing the effect of regular social audits of NREGS on programme outcomes like employment generation, income generation, community assets creation, etc. It also should be noted that Covid-19 pandemic disrupted the process of social audits in India. It will be interesting to see the program outcome in the Covid-19 years (2020 & 2021) compared to pre-Covid-19 times, controlling for other variables, as a long-term effect of social audits. The dissertation will try to study the linkage between extensiveness of the social audits and scheme deliverables at the district level.

4. *Does carrying regular social audits of MGNREGS works mandated in the Act have any impact on strengthening local democracy? Do people's knowledge about their rights and entitlements increase with the number of social audits?*

Though social audits are much talked about its participatory approach and encouraging people's democratic practice in the governance, there is very limited literature on the impact of social audit process on people and their democratic consciousness. The social audits suppose increase people's awareness about their rights and entitlements enshrined in the Act, and giving them an institutional platform to voice their issues/ grievances to authority. The dissertation intends to study the impact of social audit on people's awareness about their rights & entitlements and strengthening of local democracy, if time permits.

## **CHAPTER 4:**

### **Data and Methodology**

#### **4.1 Data**

The panel data for this study was extracted from the official website of the Society for Social Audit, Accountability and Transparency (SSAAT), Andhra Pradesh for 658 mandals across thirteen districts of Andhra Pradesh. This dissertation analyses the data at the mandal-level, rather than at the GP level. The MIS reports on the SSAAT website have eight components - audit details, expenditure and records, financial deviation, procedural deviation and delayed payment, grievances, entitlements, categorization of issues, and monthly reports. The MIS data on the SSAAT website is available from 2008 to 2022. However, our study of social audit examines five-year panel data from 2017-18 to 2021-22.

The report on the household, as well as individual-level work demand and supply, was extracted from the MIS report of the official MGNREGA website. The MGNREGA website provides mandal-level data based on the newly formed thirteen districts of Andhra Pradesh. In 2022, the state government of Andhra Pradesh notified the formation of thirteen new districts under the AP Districts Formation Act, Section 3(5). But for the purpose of this study, we have considered the list of old districts, matching with the available panel data on the SSAAT website. Hence, the new districts were renamed to their old district, from where the new districts

are carved out, to ensure consistency with the data extracted from the SSAAT website.

The control variables for the study were extracted from the 2011 Census at the district level. The latest data on education level, gender ratio, dependency ratio, urbanization level, etc. at the mandal-level is not readily available for the 2017-2022 time period. However, data on household size, total households and population density are available at the mandal level. Hence, at the mandal level and year-wise, there is less variability in control variables, which may affect the analysis of mandal level fixed effects.

## 4.2 Descriptive Statistics

Our panel data comprises of mandal level data from thirteen districts of Andhra Pradesh. The table in the appendix shows the district-wise frequency of observations and mandals. The data comprises of 3290 observations from 658 mandals of AP.

**Table 4.1: Summary of variables of social audits frequency**

| Variable                  | Obs   | Mean    | Std.<br>dev. | Min | Max |
|---------------------------|-------|---------|--------------|-----|-----|
| GPs audited once          | 2,600 | 11.0604 | 10.6095      | 0   | 46  |
| GPs audited at least once | 2,600 | 13.6315 | 10.1799      | 0   | 46  |
| GPs audited twice         | 2,600 | 2.5669  | 6.6903       | 0   | 40  |

The MGNREGA Act mandates the government to conduct social audits every 6 months, which means the study could potentially include two social audit events within a single year of the panel data, depending on the timing of the data collection. Table 4.1 presents statistics about the frequency of social audits conducted in a year. Each mandal have more than 13 GPs having conducted at

least one social audit in a year. While a few GPs, on average 2.57 GPs per mandal, have conducted two audits in a year. These variables are normalized by taking percentage of GPs audited once or twice out of total GPs.

**Table 4.2: Summary statistics of variables related to quality of SA process**

| Variable          | Obs   | Mean    | Std.<br>dev. | Min | Max    |
|-------------------|-------|---------|--------------|-----|--------|
| Total VSAs        | 2,596 | 22.5266 | 11.7994      | 0   | 82     |
| Total DRPs        | 2,596 | 11.3494 | 4.7219       | 0   | 42     |
| Total SRPs        | 2,596 | 1.4145  | 0.6195       | 1   | 5      |
| SA expenditure    | 2,596 | 106800  | 42876.21     | 0   | 302649 |
| SA met JC holders | 2,581 | 5585.92 | 2980.062     | 0   | 19279  |
| Independent jury  | 2,581 | 2.0415  | 3.7698       | 0   | 27     |

Table 4.2 represents statistics about the variables influencing quality of social audit such as human resource deployed by the government to conduct social audits, fund spent on implementation of social audits, presence of independent jury during public hearings. On average, a team of 22 village social auditors and 11 district resource persons in each mandal is deployed for conducting social audits. Two independent jury members were present during public hearings of social audits.

**Table 4.3: Summary statistics of social audit findings**

| Variable           | Obs   | Mean     | Std.<br>dev. | Min    | Max      |
|--------------------|-------|----------|--------------|--------|----------|
| Works audited      | 2,603 | 1630.391 | 1040.645     | 0      | 8155     |
| Fin. audited amt   | 2,603 | 8.17E+07 | 5.59E+07     | 641881 | 4.43E+08 |
| Social audited amt | 2,603 | 4.63E+07 | 4.20E+07     | 0      | 2.89E+08 |
| Total issues       | 2,603 | 196.627  | 641.682      | 1      | 32107    |
| Fin. deviation     | 2,603 | 4.499608 | 5.408643     | 0.05   | 56.97    |
| Recovery in SA     | 2,603 | 6866.535 | 44558.95     | 0      | 1103356  |

|                 |       |          |          |   |          |
|-----------------|-------|----------|----------|---|----------|
| Penalty imposed | 2,603 | 15.49251 | 81.54589 | 0 | 3948     |
| Issues resolved | 2,603 | 4.731848 | 13.45649 | 0 | 221      |
| Delayed wage    | 2,603 | 454059.2 | 1258529  | 0 | 1.68E+07 |

Table 4.3 presents data of social audit findings. On average, every mandal has witnessed social audits of 1630 number of works, amounting to Rs. 4.63 crores. Social audit team was able to find on overage 196 issues of misappropriation, mismanagement and corruption in audited GPs of a mandal. The financial misappropriation amounts to 4.5%, and in one case 57%, of total expenditure which is a significant proportion, given the scale of MGNREGA budget. However, merely Rs. 6866 is recovered, on average, from the financial misappropriated amount and every mandal sees only 15 cases where penalty is imposed on transgressors.

**Table 4.4: Summary statistics of variables relating to MGNREGA performance**

| Variable          | Obs   | Mean     | Std. dev. | Min     | Max     |
|-------------------|-------|----------|-----------|---------|---------|
| HH issued JC      | 3,034 | 13689.2  | 4072.107  | 3607    | 35004   |
| HH work demand    | 3,034 | 7113.991 | 3452.829  | 0       | 23797   |
| HH work supplied  | 3,034 | 6447.334 | 2933.941  | 223     | 19762   |
| Ind. work demand  | 3,034 | 10572.04 | 5225.058  | 311     | 33218   |
| Total person-days | 3,034 | 343062.4 | 209103.1  | 4601    | 1516450 |
| Avg person-days   | 3,034 | 32.16664 | 9.346394  | 10.0983 | 66.4248 |
| HH with 100 days  | 3,034 | 995.6496 | 1080.351  | 1       | 11581   |

Similarly, in table 4.4, figures of MGNREGA performance are given. In the 658 mandals of AP, on average 32 person-days of works is generated. Under MGNREGA act, every family is entitled to get 100 days of work from the state. Out of 6447 families in a mandal which were provided with at least one day of work, around 996 households were able to get work of 100 days in a year in a mandal. While, it

can be noted from the table that almost 90% households who demand work (7114 HHs) are getting some kind of work under the MGNREGA scheme.

### **4.3 Methodology**

The research design for this study is a quantitative panel data analysis of the impact of social audits on the leakages and performance of the MGNREGA scheme in the state of Andhra Pradesh. The study will use a five-year panel dataset spanning the financial years 2017-18 to 2021-22. Panel data analysis enables the study of changes within the same units of observations throughout a time period. Our study will analyse the data using three regression models: pooled OLS, fixed effects, and random effects. These models are commonly employed in panel data analysis to account for unobserved variability across the unit of observation and time. The research analysis will take place on the mandal level data of the state of Andhra Pradesh. Though the social audit process takes place at the gram panchayat level, we are considering that social audits will have mandal level effects. Because the block-level social audit team conducting audits in all GPs of a mandal is same. Hence, we can assume some uniformity in the intensity and quality of social audits in the GPs for a particular mandal. The dependent variables for our study can be loosely categorized into two types – performance variables and leakage variables. The performance variables mainly indicate the implementation quality of MGNREGA program. The variables include average person-days of work generated, number of households & individuals provided with work, households completed 100 days of work, wage expenditure, total work generated, number of job cards issued, delayed payment, etc. The other group of variables explains the degree of leakages, mismanagement and corrective action. The variables are financial deviation,

recovery done under social audit, penalty imposed upon transgressors, total issues reported during SA, issued resolved, etc.

The explanatory variables include the frequency of social audits, the number of village social auditors and district resource persons, the amount socially audited, the presence of an independent jury in public hearings, and social audit expenditure. These variables illustrate the intensity and quality of social audits taking place in mandals. The regression analysis is controlled for the variables like average household size, gender ratio, literacy ratio, urbanization rate, dependency ratio, population density, and covid dummy for the year 2020-21 and 2021-22. The data is extracted from the 2011 Census of India. The gender ratio is calculated by taking the ratio of female to male population in the district, multiplied by 1000. Similarly, literacy ratio is calculated by taking the proportion of literate population of total population. Urbanization rate is the proportion of population living in the urban areas compared to total population. For the dependency ratio, age group of 0-14 years and 60+ years population is assumed to be dependent on the working population. Hence, dependency ratio is taken by calculating dependent population as a percentage of total population. The covid dummy is a binary variable which takes values as 0 and 1. For the years 2017-18, 2018-19 and 2019-2020, value of covid dummy is 0 and it is 1 for remaining two years.

As noted earlier, we are taking panel data structure to analyze whether MGNREGA program implementation improves with repeated social audits in mandals over time, while controlling for other factors which may potentially influence program implementation quality. The regression analysis is carried out on three different regression models. The specifications of these models are given below:

Model 1:

$$Y_i = \beta_0 + \beta_1 * pc\_audited\_once_{it} + \beta_2 * pc\_audited\_twice_{it} + \beta_3 \\ * sa\_expenditure_{it} + \beta_4 * social\_audited\_amt_{it} + \beta_5 \\ * works\_audited_{it}$$

As noted earlier, the analysis is conducted at the mandal level. The dependent variable for mandal  $i$  is denoted by  $Y_i$ . The outcomes comprises of different variables relating to MGNREGA performance and leakages, as described in earlier. The main coefficients of interest,  $pc\_audited\_once$ ,  $pc\_audited\_twice$ , and  $sa\_expenditure$ , are percent of GPs audited once and twice in a given year and expenditure on social audits per GP, respectively, for mandal  $i$  and year  $t$ . The coefficients on these variables will tell us whether with frequency of social audits affect program implementation quality. In model 1, we do not take into account other confounding factors and control variables.

Model 2:

$$Y_i = \beta_0 + \beta_1 * pc\_audited\_once_{it} + \beta_2 * pc\_audited\_twice_{it} + \beta_3 \\ * sa\_expenditure_{it} + \beta_4 * social\_audited\_amt_{it} + \beta_5 \\ * works\_audited_{it} + \beta_6 * vsa_{it} + \beta_7 * drp_{it} + \beta_8 * total\_jury_{it} \\ + \beta_9 * sa\_met\_jcholders_{it} + \sum \beta_n * \mathbf{Control}$$

Model 3:

$$Y_i = \beta_0 + \beta_1 * pc\_audited\_once_{it} + \beta_2 * pc\_audited\_twice_{it} + \beta_3 \\ * sa\_expenditure_{it} + \beta_4 * social\_audited\_amt_{it} + \beta_5 \\ * works\_audited_{it} + \sum \beta_n * \mathbf{Control}$$

In model 2, we consider more variables related to social audits, which might have effect on the quality of social audits subsequently affecting our dependent variables. Hence, we add human resource variables like village social auditors and district resource persons per GP into our model. Similarly, number of independent jury present for public hearings and number of job card holders met by social audit team illustrates the quality of social audit in terms of transparency and impartiality.

Control is a vector of demographic characteristics that includes average household size, gender ratio, literacy ratio, urbanization rate, dependency ratio and population density at district level. It also includes a dummy variable for whether the year was a covid year.

To address the concern of selecting an appropriate regression model for the analysis, we undertake two tests – LM and Hausman test. The LM test, also known as the Lagrange Multiplier test, is used to test for the presence of omitted variable bias in a fixed effects regression model. If the LM test statistic is statistically significant, it suggests that the random effects model is more appropriate than the pooled OLS model. While the Hausman test is used to choose between a fixed effects and a random effects model. The test compares the coefficients of the two models and calculates the difference between them. If the difference is statistically significant, it suggests that the fixed effects model is more appropriate than the random effects model. Hence, if we get significant coefficient for LM test which means we should not consider pooled OLS model but random effects model. In Hausman test, if we get significant result which random effects model should be rejected and fixed effects model is the most appropriate model for our analysis. The table below presents the result related to Hausman and Lagrange Multiplier test conducted on model (2) with different dependent variables.

**Table 4.5: Results of Hausman and LM test**

| <b>Dependent variable</b>      | <b>Hausman test</b> | <b>LM test</b>  | <b>Model</b>       |
|--------------------------------|---------------------|-----------------|--------------------|
| Wage expenditure               | Significant         | Not significant | Fixed; Pooled OLS  |
| Total expenditure              | Significant         | Significant     | Fixed              |
| Average person-days            | Significant         | Significant     | Fixed              |
| HH completed 100 days of work  | Significant         | Significant     | Fixed              |
| HH provided with work          | Significant         | Significant     | Fixed              |
| Individuals provided with work | Significant         | Significant     | Fixed              |
| HH provided with job cards     | Significant         | Significant     | Fixed              |
| Total job cards issued         | Significant         | Significant     | Fixed              |
| Total works                    | Significant         | Significant     | Fixed              |
| Delayed wage payment           | Not significant     | Significant     | Random; Pooled OLS |
| Financial deviation            | Significant         | Significant     | Fixed              |
| Recovery done in SA            | Not significant     | Not significant | Random             |
| Penalty imposed                | Not significant     | Not significant | Random             |
| Records not submitted          | Significant         | Not significant | Fixed; Pooled OLS  |
| Total issues reported          | Not significant     | Not significant | Random             |
| Total issues resolved          | Not significant     | Not significant | Random             |

## CHAPTER 5:

### Results and Discussion

In this section, we will discuss the regression results that we run on our panel data. The regression tables present coefficients of three regression models – pooled OLS, fixed effects and random effects. The three columns in each regression model, column (1), (2) and (3), represent three different regression specifications, earlier discussed in methodology section. The specification of column (1) has only main independent variables related to social audit. While column (2) and (3) specification include control variables comprising of demographic characteristics of districts. There are additional independent variables describing quality aspect of social audit in the specification (2).

Table 5.1 presents variation in program’s wage expenditure across social audits variables, controlling for other demographic variables. The findings of the fixed effects model regression on the panel data suggest that social audit expenditure and socially audited amount have a significant positive effect on total wage expenditure of MGNREGA. Wage expenditure signifies the amount of budget went to poor families who worked under MGNREGA scheme; hence wage expenditure is one of the main variables of program implementation quality. The positive coefficient for social audit expenditure indicates that as the expenditure on social audits increases, there is a corresponding increase in total wage expenditure. This suggests that the resources allocated towards social audits can have a positive impact on program performance. Similarly, as there is an increase in social audited amount, which indicates the coverage of total expenditure by social audits, there is a significant increase in wage expenditure. The social audit team meeting job card

holders is an important indicator of increasing local participation and bringing transparency to the process. The findings in the table 5.1 suggests that the number of job card holders social audit team meets has a positive effect on wage expenditure, suggesting that with increased transparency and local participation, the program outcomes improve significantly. These findings are consistent with our theoretical framework of accountability theory. However, the number of works audited in social audit process has a significant negative relationship with our dependent variable which is counterintuitive, as one would expect that increased scrutiny of the program through social audits would lead to improved program performance and higher wage expenditure. It is possible that the social audit team may focus their auditing efforts on a particular set of works, such as those that are more likely to be associated with leakages or corruption. If these works are also associated with higher wage expenditure, then there could be a spurious negative relationship between works audited and wage expenditure. However, it is also possible that the social audit process itself may be creating disincentives for program implementation, which could be contributing to the negative relationship between works audited and wage expenditure.

In table 5.2 and 5.3, we examine the effect of social audits on average person-days of work and households who completed 100 days of work generated in the program. The findings from the fixed effects model suggest that the number of village social auditors has a positive significant coefficient with average person-days of work is consistent with the expectation that greater accountability through social audits can improve program performance and increase average person-days generated under the scheme. However, social audited amount and number of job card holders SA team met negatively affect both the dependent variables. A possible explanation is that the social audit process may be creating disincentives for program

implementation. If the social audit process is perceived as overly burdensome or adversarial, then it may create a disincentive for local officials to implement the program effectively, leading to lower average person-days of work and families completing 100 days of work. The covid year dummy variable has a strong positive coefficient with households completing 100 days of work. This means, number of households, which were provided with 100 complete days of work, were significantly higher in covid years (2020-21 & 2021-22) than non-covid years (2017-18, 2018-19 & 2019-20). Other variables don't seem to have any significant relationship with average person-days.

In table 5.4, we look at how social audits affect number of job cards issued to beneficiaries, as job cards issued indicates program quality. The result suggests that as number of works audited, village social auditors and job card holders social audit team met increase, it has significant positive effect on job cards issued to beneficiaries. The findings are consistent with the earlier results and expectation that greater accountability through social audits can increase quality of program implementation. However, coefficient of social audited amount shows a slightly small negative effect on the given dependent variable. This result should be considered for further investigation to explain potential mechanisms of the effect.

The findings of above regression analysis on program performance variables show that social audit process impacts performance of MGNREGA program through resources allotted towards social audit and increased transparency. Now, we look at how social audit process influence the leakage issue in MGNREGA scheme. In table 5.5, we investigate the variations in financial deviation as we change social audit variables, controlling for other independent variables. The findings of the regression analysis show a mixed result of social audits on financial deviation

reported. The percentage of GPs which are audited once and twice and the expenditure on social audit process have a positive relationship with the percent of financial deviation found in SA. This suggests that more frequent social audits and increased expenditure on the social audit process increase reporting of financial misappropriation. This means more rigorous social audit process will increase in exposing financial mismanagement. The result also illustrates that as social audit become more intense by increasing number of works audited, it has positive effect on reported financial misappropriation. These results are very much consistent with the accountability theory discussed in theoretical framework. As monitoring of public schemes become transparent and participatory, more cases of financial misappropriation will be uncovered. However, this public scrutiny shall eventually follow into lowering leakages and corruption. But this effect is not clear from the current analysis.

The counterintuitive negative coefficients for social audited amount and number of village social auditors are puzzling. One possible explanation could be that these variables are capturing some unobserved factors that are negatively affecting reporting of financial deviation, and not the direct effect of social audits. Another possible explanation could be that the quality of social audits conducted by some village social auditors may not be up to the mark, and there is variability in the quality of social audits, leading to confusing effect on the percent of financial deviation. Overall, these findings highlight the complex relationship between social audits and financial deviation, and suggest that policymakers should carefully consider the frequency and quality of social audits in order to effectively reduce program leakages in the MGNREGA scheme.

We do not claim the robustness of the data, as it is downloaded from the social audit unit of Andhra Pradesh, which may have some accuracy issues with data collection. Also, the demographic control variables are extracted from the 2011 Census data, and it is a district level data. Hence, control variables lack variability across mandals and yearly time period. This may affect our results and hence require a further careful inquiry. The results of Hausman and LM tests on our models were also not consistent. A few of the regression models did not give a clear answer to which regression model is appropriate for the analysis. In regression model with wage expenditure as the dependent variable, Hausman test rejected the null hypothesis indicating the use of fixed effects model. But LM test did not reject the null hypothesis which means pooled OLS as a more appropriate model. Similarly, not submitted records have similar test results. The test results of regressions with dependent variables delayed wage, recovery during SA process, penalty imposed on transgressors, number of issues reported and resolved have shown that pooled OLS model is more appropriate. The regression tables of other variables can be found in Appendix section.

**Table 5.1: Effect of social audits on total wage expenditure**

| Wage expenditure         | Pooled OLS             |                       |                        | Fixed Effects          |                       |                       | Random Effects         |                       |                        |
|--------------------------|------------------------|-----------------------|------------------------|------------------------|-----------------------|-----------------------|------------------------|-----------------------|------------------------|
|                          | (1)                    | (2)                   | (3)                    | (1)                    | (2)                   | (3)                   | (1)                    | (2)                   | (3)                    |
| pc_gp_audited_once       | -3.489<br>(-0.31)      | -15.95<br>(-1.32)     | -13.38<br>(-1.08)      | -23.30<br>(-1.32)      | -19.07<br>(-1.09)     | -22.13<br>(-1.13)     | -4.851<br>(-0.42)      | -18.68<br>(-1.49)     | -14.21<br>(-1.13)      |
| pc_gp_audited_twice      | 1.611<br>(0.15)        | -3.630<br>(-0.30)     | -6.258<br>(-0.50)      | -20.22<br>(-1.17)      | -12.11<br>(-0.70)     | -26.64<br>(-1.38)     | -0.248<br>(-0.02)      | -8.101<br>(-0.65)     | -8.470<br>(-0.67)      |
| sa_expense_percp         | 0.341***<br>(7.22)     | 0.271***<br>(3.32)    | 0.293***<br>(5.25)     | 0.458***<br>(6.98)     | 0.267**<br>(2.92)     | 0.551***<br>(7.20)    | 0.350***<br>(7.34)     | 0.271***<br>(3.36)    | 0.314***<br>(5.56)     |
| social_audited_amt_percp | 0.000689***<br>(10.82) | 0.000582***<br>(7.66) | 0.000791***<br>(10.79) | 0.000946***<br>(10.33) | 0.000590***<br>(6.35) | 0.000908***<br>(8.92) | 0.000710***<br>(10.98) | 0.000580***<br>(7.55) | 0.000803***<br>(10.81) |
| works_audited_percp      | -3.498<br>(-1.27)      | -6.836*<br>(-2.22)    | -5.495<br>(-1.78)      | -6.658<br>(-1.83)      | -10.38**<br>(-2.91)   | -8.070*<br>(-2.01)    | -3.717<br>(-1.34)      | -7.999**<br>(-2.60)   | -5.812<br>(-1.87)      |
| vsa_percp                |                        | -541.0*<br>(-2.01)    |                        |                        | -203.9<br>(-0.68)     |                       |                        | -498.6<br>(-1.88)     |                        |
| drp_percp                |                        | -733.0<br>(-1.51)     |                        |                        | -738.6<br>(-1.36)     |                       |                        | -817.8<br>(-1.71)     |                        |
| total_jury_percp         |                        | -1473.8*<br>(-2.26)   |                        |                        | -1369.6<br>(-1.67)    |                       |                        | -1640.3*<br>(-2.48)   |                        |
| sa_met_jcholders_percp   |                        | 9.296***              |                        |                        | 21.67***              |                       |                        | 11.85***              |                        |

|                  |                   |                       |                    |                       |                  |                  |                   |                       |                    |
|------------------|-------------------|-----------------------|--------------------|-----------------------|------------------|------------------|-------------------|-----------------------|--------------------|
|                  |                   | (8.61)                |                    | (16.28)               |                  |                  |                   | (10.86)               |                    |
| avg_hhsize       |                   | -526.0<br>(-1.37)     | -366.3<br>(-0.94)  | 0<br>(.)              | 0<br>(.)         |                  |                   | -508.4<br>(-1.16)     | -355.2<br>(-0.86)  |
| gender_ratio     |                   | -31.42**<br>(-2.78)   | -10.26<br>(-0.92)  | 0<br>(.)              | 0<br>(.)         |                  |                   | -36.00**<br>(-2.82)   | -10.10<br>(-0.86)  |
| literacy_rate    |                   | 47.20<br>(1.31)       | -1.890<br>(-0.05)  | 0<br>(.)              | 0<br>(.)         |                  |                   | 65.42<br>(1.61)       | -1.744<br>(-0.05)  |
| urban_rate       |                   | 9.191<br>(0.61)       | 2.899<br>(0.19)    | 0<br>(.)              | 0<br>(.)         |                  |                   | 14.03<br>(0.81)       | 3.188<br>(0.20)    |
| dependency_ratio |                   | 133.0<br>(0.79)       | -102.0<br>(-0.62)  | 0<br>(.)              | 0<br>(.)         |                  |                   | 216.1<br>(1.13)       | -107.8<br>(-0.62)  |
| popul_density    |                   | 2.194***<br>(7.21)    | 1.903***<br>(6.24) | 0<br>(.)              | 0<br>(.)         |                  |                   | 2.291***<br>(6.57)    | 1.920***<br>(5.93) |
| covid_year       |                   | -1446.9***<br>(-3.90) | -217.2<br>(-0.63)  | -1545.8***<br>(-4.00) | 893.8*<br>(2.35) |                  |                   | -1442.2***<br>(-4.03) | -100.8<br>(-0.30)  |
| _cons            | 2672.9*<br>(2.46) | 28646.3<br>(1.87)     | 17749.5<br>(1.16)  | 3719.7*<br>(2.15)     | 174.4<br>(0.10)  | 3014.4<br>(1.56) | 2738.7*<br>(2.47) | 28986.3<br>(1.66)     | 17665.0<br>(1.09)  |
| <i>N</i>         | 1716              | 1332                  | 1347               | 1716                  | 1332             | 1347             | 1716              | 1332                  | 1347               |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table 5.2: Effect of social audits on average person-days of work generated**

| avg_persondays_percapita  | Pooled OLS                  |                             |                            | Fixed Effects           |                        |                             | Random Effects            |                            |                           |
|---------------------------|-----------------------------|-----------------------------|----------------------------|-------------------------|------------------------|-----------------------------|---------------------------|----------------------------|---------------------------|
|                           | (1)                         | (2)                         | (3)                        | (1)                     | (2)                    | (3)                         | (1)                       | (2)                        | (3)                       |
| pc_gp_audited_once        | -0.0150<br>(-0.58)          | 0.00556<br>(0.20)           | 0.00271<br>(0.10)          | 0.0163<br>(0.71)        | 0.000964<br>(0.04)     | 0.00629<br>(0.24)           | 0.00349<br>(0.16)         | 0.00259<br>(0.11)          | 0.00679<br>(0.28)         |
| pc_gp_audited_twice       | -0.0300<br>(-1.19)          | 0.00451<br>(0.16)           | 0.00659<br>(0.23)          | -0.00424<br>(-0.19)     | -0.00950<br>(-0.37)    | -0.00118<br>(-0.05)         | -0.0163<br>(-0.78)        | -0.00522<br>(-0.22)        | 0.00203<br>(0.08)         |
| sa_expense_percgp         | 0.0000811<br>(0.75)         | -0.0000489<br>(-0.25)       | 0.0000732<br>(0.57)        | 0.0000378<br>(0.45)     | -0.000259<br>(-1.90)   | -0.0000911<br>(-0.89)       | 0.0000349<br>(0.42)       | -0.000233<br>(-1.71)       | -0.0000504<br>(-0.51)     |
| social_audited_amt_percgp | 0.0000017**<br>*<br>(11.75) | 0.0000018**<br>*<br>(10.13) | 0.0000016**<br>*<br>(9.58) | -0.000005***<br>(-3.85) | -0.0000003*<br>(-1.98) | -<br>0.0000004**<br>(-3.08) | 0.00000016<br>4<br>(1.43) | 0.00000027<br>7*<br>(2.04) | 0.00000013<br>5<br>(1.03) |
| works_audited_percgp      | -0.0355***<br>(-5.64)       | -0.0365***<br>(-5.04)       | -0.0391***<br>(-5.49)      | 0.00296<br>(0.63)       | 0.00500<br>(0.94)      | 0.00394<br>(0.73)           | -0.00759<br>(-1.64)       | -0.00516<br>(-0.98)        | -0.00653<br>(-1.24)       |
| vsa_percgp                |                             | 1.779**<br>(2.82)           |                            |                         | 2.009***<br>(4.49)     |                             |                           | 2.004***<br>(4.49)         |                           |
| drp_percgp                |                             | -1.143<br>(-1.00)           |                            |                         | -0.356<br>(-0.44)      |                             |                           | -0.558<br>(-0.69)          |                           |
| total_jury_percgp         |                             | 2.040<br>(1.33)             |                            |                         | 1.308<br>(1.07)        |                             |                           | 1.431<br>(1.21)            |                           |

|                        |                        |                      |                    |                     |                        |                     |                     |                        |                     |
|------------------------|------------------------|----------------------|--------------------|---------------------|------------------------|---------------------|---------------------|------------------------|---------------------|
| sa_met_jcholders_percp | -0.00847***<br>(-3.34) |                      |                    |                     | -0.00941***<br>(-4.73) |                     |                     | -0.00821***<br>(-4.24) |                     |
| avg_hhsize             | 2.192*<br>(2.44)       | 1.975*<br>(2.20)     |                    |                     | 0<br>(.)               | 0<br>(.)            |                     | 1.367<br>(1.04)        | 1.135<br>(0.86)     |
| gender_ratio           | 0.108***<br>(4.05)     | 0.0927***<br>(3.61)  |                    |                     | 0<br>(.)               | 0<br>(.)            |                     | 0.148***<br>(4.02)     | 0.130***<br>(3.55)  |
| literacy_rate          | -0.178*<br>(-2.10)     | -0.104<br>(-1.27)    |                    |                     | 0<br>(.)               | 0<br>(.)            |                     | -0.323**<br>(-2.75)    | -0.256*<br>(-2.19)  |
| urban_rate             | -0.0937**<br>(-2.64)   | -0.0937**<br>(-2.66) |                    |                     | 0<br>(.)               | 0<br>(.)            |                     | -0.135**<br>(-2.62)    | -0.134**<br>(-2.61) |
| dependency_ratio       | 0.277<br>(0.70)        | 0.496<br>(1.30)      |                    |                     | 0<br>(.)               | 0<br>(.)            |                     | 0.525<br>(0.94)        | 0.735<br>(1.33)     |
| popul_density          | 0.000385<br>(0.54)     | 0.000956<br>(1.36)   |                    |                     | 0<br>(.)               | 0<br>(.)            |                     | -0.0000612<br>(-0.06)  | 0.000437<br>(0.42)  |
| covid_year             | 0.0589<br>(0.07)       | -1.679*<br>(-2.13)   |                    |                     | 0.886<br>(1.54)        | -0.969<br>(-1.91)   |                     | 0.588<br>(1.01)        | -1.154*<br>(-2.24)  |
| _cons                  | 32.94***<br>(13.27)    | -79.52*<br>(-2.21)   | -75.42*<br>(-2.13) | 31.77***<br>(14.21) | 33.75***<br>(13.00)    | 32.80***<br>(12.72) | 32.62***<br>(15.48) | -112.8*<br>(-2.21)     | -105.3*<br>(-2.06)  |
| <i>N</i>               | 1716                   | 1332                 | 1347               | 1716                | 1332                   | 1347                | 1716                | 1332                   | 1347                |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table 5.3: Effect of social audits on number of households completing 100 days of work**

| HHs completing 100 days  | Pooled OLS              |                       |                        | Fixed Effects          |                             |                             | Random Effects        |                       |                       |
|--------------------------|-------------------------|-----------------------|------------------------|------------------------|-----------------------------|-----------------------------|-----------------------|-----------------------|-----------------------|
|                          | (1)                     | (2)                   | (3)                    | (1)                    | (2)                         | (3)                         | (1)                   | (2)                   | (3)                   |
| pc_gp_audited_once       | -0.127<br>(-0.93)       | 0.0797<br>(0.53)      | 0.121<br>(0.80)        | 0.0721<br>(0.55)       | -0.0143<br>(-0.10)          | 0.00119<br>(0.01)           | -0.00682<br>(-0.06)   | 0.0849<br>(0.60)      | 0.0976<br>(0.70)      |
| pc_gp_audited_twice      | -0.337*<br>(-2.53)      | -0.241<br>(-1.58)     | -0.233<br>(-1.52)      | -0.0918<br>(-0.73)     | -0.275<br>(-1.88)           | -0.235<br>(-1.62)           | -0.176<br>(-1.47)     | -0.187<br>(-1.34)     | -0.175<br>(-1.27)     |
| sa_expense_pergp         | -0.000151<br>(-0.27)    | -0.00149<br>(-1.45)   | 0.00137*<br>(1.99)     | -0.00161***<br>(-3.34) | -0.00122<br>(-1.57)         | -0.00104<br>(-1.80)         | -0.000946*<br>(-1.97) | -0.00140<br>(-1.71)   | 0.000118<br>(0.20)    |
| social_audited_amt_pergp | 0.0000118***<br>(15.39) | 0.00000879*<br>(9.23) | 0.0000102**<br>(11.35) | -0.00000122<br>(-1.81) | -<br>0.00000157*<br>(-1.99) | -<br>0.0000023**<br>(-2.99) | 0.00000371*<br>(5.59) | 0.00000222*<br>(2.75) | 0.00000222*<br>(2.88) |
| works_audited_pergp      | -0.0687*<br>(-2.07)     | -0.0966*<br>(-2.50)   | -0.0669<br>(-1.76)     | -0.0482<br>(-1.80)     | -0.0314<br>(-1.03)          | -0.0326<br>(-1.08)          | -0.0595*<br>(-2.20)   | -0.0476<br>(-1.51)    | -0.0377<br>(-1.22)    |
| vsa_pergp                |                         | 5.053<br>(1.50)       |                        |                        | 4.758<br>(1.87)             |                             |                       | 5.599*<br>(2.09)      |                       |
| drp_pergp                |                         | 10.37<br>(1.70)       |                        |                        | 6.916<br>(1.50)             |                             |                       | 10.02*<br>(2.06)      |                       |
| total_jury_pergp         |                         | 0.460<br>(0.06)       |                        |                        | 6.174<br>(0.89)             |                             |                       | 5.353<br>(0.76)       |                       |

|                        |                       |                       |                   |                    |                       |                    |                    |                       |                       |
|------------------------|-----------------------|-----------------------|-------------------|--------------------|-----------------------|--------------------|--------------------|-----------------------|-----------------------|
| sa_met_jcholders_pergp | 0.0678***<br>(5.01)   |                       |                   |                    | -0.0539***<br>(-4.77) |                    |                    | 0.00227<br>(0.20)     |                       |
| avg_hhsize             | 17.67***<br>(3.68)    | 18.25***<br>(3.80)    |                   |                    | 0<br>(.)              | 0<br>(.)           |                    | 13.52*<br>(2.03)      | 13.11*<br>(1.97)      |
| gender_ratio           | 0.284*<br>(2.00)      | 0.385**<br>(2.80)     |                   |                    | 0<br>(.)              | 0<br>(.)           |                    | 0.500**<br>(2.66)     | 0.460*<br>(2.48)      |
| literacy_rate          | -0.994*<br>(-2.20)    | -1.297**<br>(-2.94)   |                   |                    | 0<br>(.)              | 0<br>(.)           |                    | -2.268***<br>(-3.78)  | -2.239***<br>(-3.78)  |
| urban_rate             | 0.454*<br>(2.39)      | 0.347<br>(1.84)       |                   |                    | 0<br>(.)              | 0<br>(.)           |                    | 0.103<br>(0.40)       | 0.0643<br>(0.25)      |
| dependency_ratio       | 1.907<br>(0.90)       | -0.547<br>(-0.27)     |                   |                    | 0<br>(.)              | 0<br>(.)           |                    | -0.0954<br>(-0.03)    | -0.535<br>(-0.19)     |
| popul_density          | -0.0176***<br>(-4.61) | -0.0175***<br>(-4.67) |                   |                    | 0<br>(.)              | 0<br>(.)           |                    | -0.0227***<br>(-4.31) | -0.0214***<br>(-4.08) |
| covid_year             | 12.22**<br>(2.63)     | 19.29***<br>(4.57)    |                   |                    | 17.28***<br>(5.27)    | 9.847***<br>(3.45) |                    | 14.34***<br>(4.07)    | 13.22***<br>(4.29)    |
| _cons                  | 52.40***<br>(4.01)    | -342.8<br>(-1.79)     | -343.5<br>(-1.82) | 64.40***<br>(5.08) | 70.87***<br>(4.80)    | 66.25***<br>(4.56) | 59.67***<br>(4.95) | -367.1<br>(-1.41)     | -309.8<br>(-1.20)     |
| <i>N</i>               | 1716                  | 1332                  | 1347              | 1716               | 1332                  | 1347               | 1716               | 1332                  | 1347                  |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table 5.4: Effect of social audits on total job cards issued**

| Total job cards issued       | Pooled OLS                  |                              |                        | Fixed Effects          |                             |                           | Random Effects              |                              |                        |
|------------------------------|-----------------------------|------------------------------|------------------------|------------------------|-----------------------------|---------------------------|-----------------------------|------------------------------|------------------------|
|                              | (1)                         | (2)                          | (3)                    | (1)                    | (2)                         | (3)                       | (1)                         | (2)                          | (3)                    |
| pc_gp_audited_once           | 1.024<br>(1.59)             | 0.888<br>(1.37)              | 1.575*<br>(2.16)       | 0.635<br>(0.78)        | 0.336<br>(0.35)             | 0.238<br>(0.24)           | 1.162<br>(1.79)             | 1.001<br>(1.49)              | 1.489*<br>(1.98)       |
| pc_gp_audited_twice          | 0.814<br>(1.30)             | 0.567<br>(0.87)              | 0.671<br>(0.92)        | -0.169<br>(-0.21)      | -0.205<br>(-0.22)           | -0.623<br>(-0.63)         | 0.716<br>(1.13)             | 0.642<br>(0.96)              | 0.596<br>(0.79)        |
| sa_expense_percp             | 0.0442***<br>(16.25)        | 0.00340<br>(0.77)            | 0.0449***<br>(13.62)   | 0.0217***<br>(7.02)    | -0.0000937<br>(-0.02)       | 0.0198***<br>(5.08)       | 0.0354***<br>(13.47)        | 0.00295<br>(0.68)            | 0.0370***<br>(11.39)   |
| social_audited_amt_per<br>gp | -<br>0.00000784*<br>(-2.18) | -<br>0.0000272***<br>(-6.77) | -0.00000497<br>(-1.16) | -0.00000575<br>(-1.37) | -<br>0.00000994*<br>(-2.01) | 0.00000046<br>5<br>(0.09) | -<br>0.00000775*<br>(-2.19) | -<br>0.0000237***<br>(-5.91) | -0.00000352<br>(-0.83) |
| works_audited_percp          | 1.458***<br>(9.39)          | 1.092***<br>(6.69)           | 1.444***<br>(7.96)     | 0.838***<br>(4.97)     | 0.787***<br>(4.10)          | 0.873***<br>(4.36)        | 1.238***<br>(8.37)          | 1.067***<br>(6.59)           | 1.289***<br>(7.37)     |
| vsa_percp                    |                             | 70.80***<br>(4.93)           |                        |                        | 40.31*<br>(2.47)            |                           |                             | 65.82***<br>(4.66)           |                        |
| drp_percp                    |                             | 152.3***<br>(5.88)           |                        |                        | 54.35<br>(1.83)             |                           |                             | 135.5***<br>(5.31)           |                        |
| total_jury_percp             |                             | 1.944<br>(0.06)              |                        |                        | 53.16<br>(1.17)             |                           |                             | 16.93<br>(0.48)              |                        |

|                         |                      |                      |                     |                    |                    |                    |                      |                      |                     |
|-------------------------|----------------------|----------------------|---------------------|--------------------|--------------------|--------------------|----------------------|----------------------|---------------------|
| sa_met_jcholders_percgp | 1.000***<br>(17.55)  |                      | 0.646***<br>(8.98)  |                    |                    |                    | 0.938***<br>(16.37)  |                      |                     |
| avg_hhsize              | 61.20**<br>(2.96)    | 67.56**<br>(2.91)    | 0<br>(.)            | 0<br>(.)           |                    |                    | 60.82**<br>(2.61)    | 63.68*<br>(2.24)     |                     |
| gender_ratio            | -5.484***<br>(-9.31) | -3.923***<br>(-6.11) | 0<br>(.)            | 0<br>(.)           |                    |                    | -5.679***<br>(-8.69) | -4.474***<br>(-5.77) |                     |
| literacy_rate           | -2.209<br>(-1.15)    | -7.243***<br>(-3.42) | 0<br>(.)            | 0<br>(.)           |                    |                    | -2.414<br>(-1.14)    | -7.209**<br>(-2.84)  |                     |
| urban_rate              | -3.548***<br>(-4.48) | -4.974***<br>(-5.61) | 0<br>(.)            | 0<br>(.)           |                    |                    | -3.615***<br>(-4.07) | -5.112***<br>(-4.73) |                     |
| dependency_ratio        | -49.27***<br>(-5.64) | -84.16***<br>(-8.81) | 0<br>(.)            | 0<br>(.)           |                    |                    | -51.75***<br>(-5.31) | -86.61***<br>(-7.47) |                     |
| popul_density           | 0.0215<br>(1.33)     | 0.0172<br>(0.96)     | 0<br>(.)            | 0<br>(.)           |                    |                    | 0.0230<br>(1.27)     | 0.0220<br>(1.00)     |                     |
| covid_year              | -35.07<br>(-1.78)    | 68.63***<br>(3.43)   | -40.89<br>(-1.96)   | 14.08<br>(0.74)    |                    |                    | -36.62<br>(-1.93)    | 49.57**<br>(2.75)    |                     |
| _cons                   | 222.9***<br>(3.60)   | 7056.3***<br>(8.90)  | 6973.5***<br>(7.89) | 430.7***<br>(5.37) | 328.9***<br>(3.44) | 460.8***<br>(4.67) | 275.0***<br>(4.35)   | 7360.6***<br>(8.32)  | 7675.6***<br>(7.16) |
| <i>N</i>                | 1815                 | 1423                 | 1423                | 1815               | 1423               | 1423               | 1815                 | 1423                 | 1423                |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table 5.5: Effect of social audits on financial deviation issued in SA**

| Financial deviation      | Pooled OLS             |                       |                      | Fixed Effects         |                        |                       | Random Effects        |                       |                      |
|--------------------------|------------------------|-----------------------|----------------------|-----------------------|------------------------|-----------------------|-----------------------|-----------------------|----------------------|
|                          | (1)                    | (2)                   | (3)                  | (1)                   | (2)                    | (3)                   | (1)                   | (2)                   | (3)                  |
| pc_gp_audited_once       | -0.0484***<br>(-3.38)  | 0.00105<br>(0.07)     | -0.00112<br>(-0.07)  | 0.0304<br>(1.50)      | 0.0819**<br>(3.29)     | 0.0724**<br>(2.90)    | -0.0304*<br>(-2.07)   | 0.00646<br>(0.40)     | 0.00389<br>(0.24)    |
| pc_gp_audited_twice      | -0.0528***<br>(-3.77)  | -0.00662<br>(-0.42)   | -0.00664<br>(-0.42)  | 0.0203<br>(1.03)      | 0.0623*<br>(2.53)      | 0.0543*<br>(2.20)     | -0.0368*<br>(-2.57)   | -0.00266<br>(-0.17)   | -0.00313<br>(-0.19)  |
| sa_expense_pergp         | 0.000254***<br>(4.19)  | 0.000466***<br>(4.40) | 0.000195**<br>(2.73) | 0.000295***<br>(3.88) | 0.000578***<br>(4.47)  | 0.000369***<br>(3.82) | 0.000257***<br>(4.27) | 0.000478***<br>(4.53) | 0.000208**<br>(2.89) |
| social_audited_amt_pergp | -0.00000015<br>(-1.87) | -<br>(-2.62)          | -<br>(-2.09)         | -0.0000002<br>(-1.91) | -0.0000003*<br>(-2.33) | -<br>(-2.40)          | -<br>(-1.99)          | -<br>(-2.62)          | -<br>(-2.17)         |
| works_audited_pergp      | 0.00379<br>(1.10)      | 0.00501<br>(1.27)     | 0.00811*<br>(2.07)   | 0.00804<br>(1.93)     | 0.0107*<br>(2.16)      | 0.0116*<br>(2.33)     | 0.00469<br>(1.37)     | 0.00540<br>(1.37)     | 0.00820*<br>(2.09)   |
| vsa_pergp                |                        | -2.311***<br>(-6.66)  |                      |                       | -2.233***<br>(-5.29)   |                       |                       | -2.305***<br>(-6.67)  |                      |
| drp_pergp                |                        | 1.851**<br>(2.96)     |                      |                       | 2.440**<br>(3.17)      |                       |                       | 1.871**<br>(3.00)     |                      |
| total_jury_pergp         |                        | -0.211<br>(-0.25)     |                      |                       | -1.593<br>(-1.36)      |                       |                       | -0.344<br>(-0.40)     |                      |

|                        |             |             |         |          |         |         |          |            |             |
|------------------------|-------------|-------------|---------|----------|---------|---------|----------|------------|-------------|
| sa_met_jcholders_pergp | 0.00293*    |             |         |          | 0.00159 |         |          | 0.00272*   |             |
|                        | (2.13)      |             |         |          | (0.85)  |         |          | (1.96)     |             |
| avg_hhsize             | 0.619       | 0.844       |         |          | 0       | 0       |          | 0.641      | 0.868       |
|                        | (1.24)      | (1.67)      |         |          | (.)     | (.)     |          | (1.21)     | (1.62)      |
| gender_ratio           | -0.0727***  | -0.0667***  |         |          | 0       | 0       |          | -0.0707*** | -0.0653***  |
|                        | (-5.11)     | (-4.79)     |         |          | (.)     | (.)     |          | (-4.71)    | (-4.44)     |
| literacy_rate          | -0.0115     | -0.0441     |         |          | 0       | 0       |          | -0.0126    | -0.0467     |
|                        | (-0.25)     | (-0.97)     |         |          | (.)     | (.)     |          | (-0.26)    | (-0.97)     |
| urban_rate             | -0.0209     | -0.0153     |         |          | 0       | 0       |          | -0.0200    | -0.0144     |
|                        | (-1.09)     | (-0.80)     |         |          | (.)     | (.)     |          | (-0.99)    | (-0.71)     |
| dependency_ratio       | 0.378       | 0.325       |         |          | 0       | 0       |          | 0.377      | 0.325       |
|                        | (1.79)      | (1.58)      |         |          | (.)     | (.)     |          | (1.69)     | (1.50)      |
| popul_density          | -0.00137*** | -0.00163*** |         |          | 0       | 0       |          | -0.00136** | -0.00162*** |
|                        | (-3.49)     | (-4.18)     |         |          | (.)     | (.)     |          | (-3.27)    | (-3.92)     |
| covid_year             | -1.297**    | -0.0165     |         |          | -0.476  | 0.857   |          | -1.188*    | 0.0837      |
|                        | (-2.73)     | (-0.04)     |         |          | (-0.88) | (1.80)  |          | (-2.54)    | (0.20)      |
| _cons                  | 7.889***    | 63.62***    | 60.30** | -0.00987 | -5.300* | -4.647  | 6.134*** | 60.97**    | 58.43**     |
|                        | (5.71)      | (3.32)      | (3.16)  | (-0.00)  | (-2.14) | (-1.89) | (4.30)   | (3.01)     | (2.89)      |
| <i>N</i>               | 1837        | 1423        | 1438    | 1837     | 1423    | 1438    | 1837     | 1423       | 1438        |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

## CHAPTER 6:

### Conclusion

As discussed earlier in literature review, social audit process through increasing transparency and accountability of government officials increases program performance (Swain & Sen, 2009; Sinha, 2008; Gurung et al, 2019; Afridi & Iversen, 2014) and keep a check on irregularities (Tambe et al., 2016; Afridi & Iversen, 2014). Social audit is a tool to democratize the public policy space to the grassroots level. Through the participation of public, i.e., beneficiaries of public schemes, and dissemination of information to the last mile, the least powerful stakeholder gains some voice in the program implementation. The policy makers who are willing to improve governance should encourage more and more social audits, not only MGNREGA but also other public schemes as well. In the light of current study, we can see that social audit positively impacts program performance through social audit expenditure, social audited amount, number of job card holders SA team met, number of village social auditors and covid dummy. There are some relationships which are puzzling and difficult to explain. We think that there must be some unobserved parameters or mechanisms which might explain the potential connection between these variables. The mixed results can also be attributed to the robustness of data.

It is noted that the frequency of social audits does not affect the performance of the MGNREGA program directly, but it can be seen that social audit expenditure has a better explaining power than the frequency variable. The social audit expenditure is in a way proportional to the social audits' frequency, which could explain the

mechanism of frequency of social audit influencing program outcomes through SA expenditure.

Based on the implications of this study, it is recommended that social audits be conducted more frequently and effectively to ensure greater transparency and accountability in the implementation of MGNREGA scheme. In addition, more attention needs to be given to the training and capacity building of social audit teams, particularly in the areas of data management and analysis. As the number of village social auditors and district resource persons affect program performance, the skilled and more efficient human resource will improve the quality of social audit, subsequently positively affect the program performance. It should be noted that as social audit teams increase, the number of beneficiaries interviewed by SA team for social audits will also increase. We can imply that SA teams meeting more job card holders will educate them about rights and entitlements and make the process more participatory and transparent. The global pandemic halted every aspect of life, especially livelihoods of millions were destroyed. However, MGNREGA emerged as a saviour for families who lost their livelihoods during covid pandemic. The government of India with increased budget to MGNREGA resulted into more wage expenditure, person-days of work, families getting work, etc. Hence, it also needs to be considered that program performance can be improved by increased budget, as was in case in covid years.

It is noted from the study findings that social audits increase the exposé of financial misappropriation. Though social audit frequency and quality will increase the uncovering corruption cases, the follow-up and enforcement process are still not democratized and the decision making remains with government officials who are under scrutiny. The social audit team or the *gram sabhas* need to be empowered for

taking disciplinary action on offenders. However, if unchecked, there may be cases of politically motivated convictions to settle scores with a person.

Andhra Pradesh has strong political will and commitment to implement social audits, however this cannot be said about other states. Hence, the social audit findings from our study cannot be generalized to other states. But we can draw some broad conclusions about the effect of social audits on program performance and leakages that will guide the general policy about accountability of public schemes.

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## Appendix

**Table A.1: Effect of social audits on total program expenditure**

| Total expenditure per capita | Pooled OLS            |                       |                       | Fixed Effects        |                       |                      | Random Effects        |                       |                      |
|------------------------------|-----------------------|-----------------------|-----------------------|----------------------|-----------------------|----------------------|-----------------------|-----------------------|----------------------|
|                              | (1)                   | (2)                   | (3)                   | (1)                  | (2)                   | (3)                  | (1)                   | (2)                   | (3)                  |
| pc_gp_audited_once           | -0.462<br>(-0.02)     | -35.47<br>(-1.65)     | -31.33<br>(-1.45)     | -49.92<br>(-1.47)    | -50.65<br>(-1.69)     | -54.60<br>(-1.68)    | -7.092<br>(-0.30)     | -42.06<br>(-1.89)     | -35.19<br>(-1.58)    |
| pc_gp_audited_twice          | -9.573<br>(-0.43)     | -20.03<br>(-0.93)     | -21.50<br>(-0.99)     | -66.75*<br>(-2.03)   | -46.32<br>(-1.56)     | -65.87*<br>(-2.05)   | -18.75<br>(-0.82)     | -30.78<br>(-1.38)     | -30.28<br>(-1.36)    |
| sa_expense_pergp             | 0.828***<br>(8.67)    | 0.424**<br>(2.91)     | 0.574***<br>(5.88)    | 0.937***<br>(7.48)   | 0.448**<br>(2.85)     | 0.931***<br>(7.32)   | 0.847***<br>(8.80)    | 0.435**<br>(3.07)     | 0.631***<br>(6.42)   |
| social_audited_amt_pergp     | 0.000600***<br>(4.67) | 0.000739***<br>(5.46) | 0.000951***<br>(7.44) | 0.00136***<br>(7.78) | 0.000798***<br>(4.99) | 0.00129***<br>(7.64) | 0.000724***<br>(5.53) | 0.000740***<br>(5.47) | 0.00101***<br>(7.82) |
| works_audited_pergp          | 2.809<br>(0.50)       | -1.398<br>(-0.25)     | 0.409<br>(0.08)       | -6.632<br>(-0.95)    | -9.752<br>(-1.59)     | -6.087<br>(-0.91)    | 1.518<br>(0.27)       | -4.174<br>(-0.77)     | -0.908<br>(-0.17)    |
| vsa_pergp                    |                       | -65.22<br>(-0.14)     |                       |                      | 64.57<br>(0.12)       |                      |                       | -90.44<br>(-0.19)     |                      |
| drp_pergp                    |                       | -105.8<br>(-0.12)     |                       |                      | -974.4<br>(-1.04)     |                      |                       | -487.7<br>(-0.58)     |                      |
| total_jury_pergp             |                       | -1073.0<br>(-0.92)    |                       |                      | 95.27<br>(0.07)       |                      |                       | -1108.8<br>(-0.95)    |                      |

|                        |                       |                       |                   |                   |                       |                    |                   |                       |                       |
|------------------------|-----------------------|-----------------------|-------------------|-------------------|-----------------------|--------------------|-------------------|-----------------------|-----------------------|
| sa_met_jcholders_pergp | 9.421***<br>(4.90)    |                       |                   |                   | 31.80***<br>(13.86)   |                    |                   | 14.88***<br>(7.73)    |                       |
| avg_hhsize             | -874.7<br>(-1.28)     | -762.4<br>(-1.12)     |                   |                   | 0<br>(.)              | 0<br>(.)           |                   | -851.6<br>(-1.07)     | -718.3<br>(-0.95)     |
| gender_ratio           | -64.07**<br>(-3.18)   | -44.97*<br>(-2.31)    |                   |                   | 0<br>(.)              | 0<br>(.)           |                   | -74.91**<br>(-3.25)   | -44.31*<br>(-2.06)    |
| literacy_rate          | 136.1*<br>(2.12)      | 88.28<br>(1.41)       |                   |                   | 0<br>(.)              | 0<br>(.)           |                   | 175.1*<br>(2.38)      | 93.94<br>(1.36)       |
| urban_rate             | 26.55<br>(0.99)       | 19.32<br>(0.72)       |                   |                   | 0<br>(.)              | 0<br>(.)           |                   | 36.73<br>(1.17)       | 20.90<br>(0.70)       |
| dependency_ratio       | 230.6<br>(0.76)       | -25.18<br>(-0.09)     |                   |                   | 0<br>(.)              | 0<br>(.)           |                   | 418.4<br>(1.21)       | -33.31<br>(-0.10)     |
| popul_density          | 5.242***<br>(9.67)    | 5.002***<br>(9.39)    |                   |                   | 0<br>(.)              | 0<br>(.)           |                   | 5.423***<br>(8.60)    | 5.052***<br>(8.50)    |
| covid_year             | -3502.9***<br>(-5.30) | -2383.8***<br>(-3.99) |                   |                   | -3848.0***<br>(-5.79) | -522.9<br>(-0.83)  |                   | -3521.6***<br>(-5.62) | -1990.7***<br>(-3.45) |
| _cons                  | 4720.6*<br>(2.15)     | 57220.0*<br>(2.10)    | 49280.2<br>(1.84) | 8226.7*<br>(2.49) | 4146.1<br>(1.38)      | 8578.2**<br>(2.67) | 5159.8*<br>(2.28) | 59000.4<br>(1.88)     | 48308.5<br>(1.63)     |
| <i>N</i>               | 1716                  | 1332                  | 1347              | 1716              | 1332                  | 1347               | 1716              | 1332                  | 1347                  |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table A.2: Effect of social audits on number of individuals provided with work**

| Individuals work provided | Pooled OLS              |                       |                        | Fixed Effects           |                      |                      | Random Effects         |                      |                       |
|---------------------------|-------------------------|-----------------------|------------------------|-------------------------|----------------------|----------------------|------------------------|----------------------|-----------------------|
|                           | (1)                     | (2)                   | (3)                    | (1)                     | (2)                  | (3)                  | (1)                    | (2)                  | (3)                   |
| pc_gp_audited_once        | -0.117<br>(-0.19)       | 0.904<br>(1.55)       | 1.326*<br>(2.04)       | 0.543<br>(1.55)         | 0.539<br>(1.47)      | 0.475<br>(1.31)      | 0.507<br>(1.41)        | 0.862*<br>(2.10)     | 0.703<br>(1.86)       |
| pc_gp_audited_twice       | -0.0813<br>(-0.13)      | -0.501<br>(-0.86)     | -0.533<br>(-0.82)      | 1.048**<br>(3.08)       | -0.0410<br>(-0.11)   | -0.0893<br>(-0.25)   | 0.999**<br>(2.85)      | 0.187<br>(0.46)      | 0.0245<br>(0.07)      |
| sa_expense_pergp          | 0.00317<br>(1.20)       | -0.00656<br>(-1.66)   | 0.0180***<br>(6.13)    | -0.00853***<br>(-6.59)  | -0.00184<br>(-0.95)  | -0.00141<br>(-1.00)  | -0.00687***<br>(-5.08) | -0.00219<br>(-0.98)  | 0.00125<br>(0.84)     |
| social_audited_amt_pergp  | 0.0000420***<br>(11.82) | 0.00000890*<br>(2.43) | 0.0000289***<br>(7.51) | 0.00000768***<br>(4.24) | 0.00000130<br>(0.67) | 0.00000186<br>(0.99) | 0.0000115***<br>(6.12) | 0.00000254<br>(1.13) | 0.00000451*<br>(2.26) |
| works_audited_pergp       | 0.464**<br>(3.02)       | 0.0537<br>(0.36)      | 0.380*<br>(2.34)       | -0.107<br>(-1.49)       | -0.0706<br>(-0.94)   | -0.0494<br>(-0.67)   | -0.0482<br>(-0.64)     | -0.0163<br>(-0.19)   | 0.0131<br>(0.16)      |
| vsa_pergp                 |                         | 17.48<br>(1.35)       |                        |                         | -4.647<br>(-0.74)    |                      |                        | 0.168<br>(0.02)      |                       |
| drp_pergp                 |                         | 57.33*<br>(2.45)      |                        |                         | 20.50<br>(1.79)      |                      |                        | 32.05*<br>(2.40)     |                       |
| total_jury_pergp          |                         | 45.86<br>(1.46)       |                        |                         | 26.63<br>(1.54)      |                      |                        | 39.08*<br>(1.97)     |                       |
| sa_met_jcholders_pergp    |                         | 0.901***              |                        |                         | 0.0224               |                      |                        | 0.192***             |                       |

|                  |                       |                      |                   |                     |                     |                     |                      |                      |                    |
|------------------|-----------------------|----------------------|-------------------|---------------------|---------------------|---------------------|----------------------|----------------------|--------------------|
|                  | (17.31)               |                      |                   | (0.80)              |                     |                     | (5.94)               |                      |                    |
| avg_hhsize       | 99.80***<br>(5.40)    | 109.5***<br>(5.35)   |                   | 0<br>(.)            | 0<br>(.)            |                     | 87.76**<br>(3.13)    | 87.84**<br>(2.76)    |                    |
| gender_ratio     | -1.322*<br>(-2.42)    | 0.341<br>(0.58)      |                   | 0<br>(.)            | 0<br>(.)            |                     | -0.548<br>(-0.70)    | -0.335<br>(-0.38)    |                    |
| literacy_rate    | -6.214***<br>(-3.57)  | -10.48***<br>(-5.58) |                   | 0<br>(.)            | 0<br>(.)            |                     | -12.62***<br>(-5.06) | -13.69***<br>(-4.89) |                    |
| urban_rate       | -0.614<br>(-0.84)     | -1.908*<br>(-2.37)   |                   | 0<br>(.)            | 0<br>(.)            |                     | -2.674*<br>(-2.44)   | -3.125*<br>(-2.52)   |                    |
| dependency_ratio | -25.12**<br>(-3.08)   | -59.62***<br>(-6.87) |                   | 0<br>(.)            | 0<br>(.)            |                     | -57.33***<br>(-4.83) | -66.76***<br>(-5.01) |                    |
| popul_density    | -0.150***<br>(-10.20) | -0.159***<br>(-9.93) |                   | 0<br>(.)            | 0<br>(.)            |                     | -0.173***<br>(-7.79) | -0.174***<br>(-6.94) |                    |
| covid_year       | 124.3***<br>(6.94)    | 225.9***<br>(12.54)  |                   | 115.8***<br>(14.22) | 120.1***<br>(17.15) |                     | 114.3***<br>(11.93)  | 130.6***<br>(17.21)  |                    |
| __cons           | 425.9***<br>(7.02)    | 2312.1**<br>(3.13)   | 2035.2*<br>(2.52) | 524.9***<br>(15.36) | 472.7***<br>(12.90) | 486.6***<br>(13.66) | 502.8***<br>(13.93)  | 3252.8**<br>(3.00)   | 3470.5**<br>(2.83) |
| <i>N</i>         | 1716                  | 1332                 | 1347              | 1716                | 1332                | 1347                | 1716                 | 1332                 | 1347               |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table A.3: Effect of social audits on number of HHs issued with job cards**

| HHs issued job cards     | Pooled OLS             |                          |                        | Fixed Effects          |                         |                         | Random Effects       |                        |                        |
|--------------------------|------------------------|--------------------------|------------------------|------------------------|-------------------------|-------------------------|----------------------|------------------------|------------------------|
|                          | (1)                    | (2)                      | (3)                    | (1)                    | (2)                     | (3)                     | (1)                  | (2)                    | (3)                    |
| pc_gp_audited_once       | 0.747<br>(1.13)        | 1.281<br>(1.89)          | 1.718*<br>(2.40)       | -0.178<br>(-0.75)      | -0.172<br>(-0.65)       | -0.220<br>(-0.83)       | -0.0731<br>(-0.30)   | 0.0740<br>(0.26)       | -0.0212<br>(-0.08)     |
| pc_gp_audited_twice      | 1.333*<br>(2.05)       | 0.709<br>(1.04)          | 0.888<br>(1.24)        | 0.363<br>(1.58)        | -0.144<br>(-0.55)       | -0.167<br>(-0.64)       | 0.484*<br>(2.04)     | 0.0822<br>(0.29)       | -0.00196<br>(-0.01)    |
| sa_expense_pergp         | 0.0242***<br>(8.72)    | 0.00453<br>(0.99)        | 0.0349***<br>(10.79)   | -0.00355***<br>(-4.06) | 0.000953<br>(0.69)      | -0.0000901<br>(-0.09)   | -0.00218*<br>(-2.40) | 0.00142<br>(0.93)      | 0.00172<br>(1.57)      |
| social_audited_amt_pergp | -0.00000293<br>(-0.78) | -0.0000191***<br>(-4.49) | -0.00000654<br>(-1.54) | 0.00000191<br>(1.57)   | -0.000000227<br>(-0.16) | -0.000000791<br>(-0.57) | 0.00000154<br>(1.22) | -0.00000140<br>(-0.91) | -0.00000119<br>(-0.82) |
| works_audited_pergp      | 1.067***<br>(6.58)     | 0.526**<br>(3.04)        | 0.872***<br>(4.87)     | -0.0219<br>(-0.45)     | -0.0250<br>(-0.46)      | -0.00367<br>(-0.07)     | 0.0296<br>(0.58)     | 0.0240<br>(0.40)       | 0.0463<br>(0.80)       |
| vsa_pergp                |                        | 46.31**<br>(3.07)        |                        |                        | -11.09*<br>(-2.45)      |                         |                      | -7.643<br>(-1.53)      |                        |
| drp_pergp                |                        | 165.0***<br>(6.05)       |                        |                        | 1.835<br>(0.22)         |                         |                      | 12.00<br>(1.33)        |                        |
| total_jury_pergp         |                        | 16.93<br>(0.46)          |                        |                        | -8.841<br>(-0.71)       |                         |                      | -3.999<br>(-0.29)      |                        |
| sa_met_jcholders_pergp   |                        | 0.602***                 |                        |                        | 0.0122                  |                         |                      | 0.0506*                |                        |

|                  |                    |                      |                       |                     |                     |                     |                     |                      |                      |
|------------------|--------------------|----------------------|-----------------------|---------------------|---------------------|---------------------|---------------------|----------------------|----------------------|
|                  |                    | (9.94)               |                       |                     | (0.61)              |                     |                     | (2.29)               |                      |
| avg_hhsize       |                    | 118.1***<br>(5.49)   | 120.2***<br>(5.33)    |                     | 0<br>(.)            | 0<br>(.)            |                     | 105.4**<br>(3.02)    | 106.5**<br>(2.90)    |
| gender_ratio     |                    | -6.046**<br>(-9.53)  | -5.358**<br>(-8.29)   |                     | 0<br>(.)            | 0<br>(.)            |                     | -7.587***<br>(-7.89) | -7.514***<br>(-7.44) |
| literacy_rate    |                    | -4.060*<br>(-2.01)   | -7.467***<br>(-3.61)  |                     | 0<br>(.)            | 0<br>(.)            |                     | -6.498*<br>(-2.12)   | -6.922*<br>(-2.15)   |
| urban_rate       |                    | -4.935***<br>(-5.82) | -5.909***<br>(-6.67)  |                     | 0<br>(.)            | 0<br>(.)            |                     | -6.403***<br>(-4.70) | -6.501***<br>(-4.54) |
| dependency_ratio |                    | -77.36***<br>(-8.15) | -102.5***<br>(-10.71) |                     | 0<br>(.)            | 0<br>(.)            |                     | -109.3***<br>(-7.45) | -111.4***<br>(-7.24) |
| popul_density    |                    | -0.0237<br>(-1.39)   | -0.0254<br>(-1.44)    |                     | 0<br>(.)            | 0<br>(.)            |                     | -0.0195<br>(-0.71)   | -0.0190<br>(-0.66)   |
| covid_year       |                    | 105.7***<br>(5.08)   | 175.1***<br>(8.82)    |                     | 48.79***<br>(8.36)  | 56.05***<br>(10.87) |                     | 51.16***<br>(7.93)   | 61.09***<br>(11.15)  |
| _cons            | 454.5***<br>(7.11) | 8599.2***<br>(10.01) | 8964.2***<br>(10.07)  | 761.3***<br>(33.05) | 740.3***<br>(28.18) | 739.2***<br>(28.16) | 734.1***<br>(28.58) | 11798.1***<br>(8.79) | 11828.3***<br>(8.38) |
| <i>N</i>         | 1716               | 1332                 | 1347                  | 1716                | 1332                | 1347                | 1716                | 1332                 | 1347                 |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table A.4: Effect of social audits on total works generated**

| Total works              | Pooled OLS                    |                                |                               | Fixed Effects                 |                                |                               | Random Effects                |                                |                               |
|--------------------------|-------------------------------|--------------------------------|-------------------------------|-------------------------------|--------------------------------|-------------------------------|-------------------------------|--------------------------------|-------------------------------|
|                          | (1)                           | (2)                            | (3)                           | (1)                           | (2)                            | (3)                           | (1)                           | (2)                            | (3)                           |
| pc_gp_audited_once       | 0.0282<br>(0.27)              | 0.0688<br>(0.67)               | 0.144<br>(1.21)               | -0.229<br>(-1.44)             | -0.222<br>(-1.37)              | -0.250<br>(-1.34)             | 0.0196<br>(0.18)              | 0.0645<br>(0.63)               | 0.142<br>(1.20)               |
| pc_gp_audited_twice      | 0.0485<br>(0.47)              | 0.0684<br>(0.67)               | 0.0361<br>(0.30)              | -0.172<br>(-1.11)             | -0.140<br>(-0.87)              | -0.282<br>(-1.54)             | 0.0449<br>(0.43)              | 0.0671<br>(0.65)               | 0.0354<br>(0.30)              |
| sa_expense_pergp         | 0.00598***<br>(13.35)         | 0.00369***<br>(5.34)           | 0.00695***<br>(13.07)         | 0.00559***<br>(9.37)          | 0.00342***<br>(4.07)           | 0.00650***<br>(9.02)          | 0.00602***<br>(13.36)         | 0.00369***<br>(5.35)           | 0.00695***<br>(13.07)         |
| social_audited_amt_pergp | -<br>0.00000524***<br>(-8.85) | -<br>0.00000942***<br>(-14.94) | -<br>0.00000506***<br>(-7.31) | -<br>0.00000407***<br>(-4.99) | -<br>0.00000842***<br>(-10.15) | -<br>0.00000540***<br>(-5.78) | -<br>0.00000518***<br>(-8.67) | -<br>0.00000941***<br>(-14.89) | -<br>0.00000506***<br>(-7.31) |
| works_audited_pergp      | 1.000***<br>(39.13)           | 0.890***<br>(34.72)            | 0.943***<br>(32.28)           | 0.726***<br>(22.19)           | 0.701***<br>(21.72)            | 0.727***<br>(19.53)           | 0.980***<br>(38.19)           | 0.884***<br>(34.45)            | 0.942***<br>(32.22)           |
| vsa_pergp                |                               | -3.944<br>(-1.75)              |                               |                               | -3.071<br>(-1.12)              |                               |                               | -3.869<br>(-1.71)              |                               |
| drp_pergp                |                               | 2.011<br>(0.49)                |                               |                               | -3.296<br>(-0.66)              |                               |                               | 1.937<br>(0.48)                |                               |
| total_jury_pergp         |                               | -7.392<br>(-1.33)              |                               |                               | -23.71**<br>(-3.11)            |                               |                               | -7.903<br>(-1.42)              |                               |
| sa_met_jcholders_pergp   |                               | 0.201***                       |                               |                               | 0.217***                       |                               |                               | 0.202***                       |                               |

|                  |        |           |           |          |           |          |        |           |           |
|------------------|--------|-----------|-----------|----------|-----------|----------|--------|-----------|-----------|
|                  |        | (22.43)   |           |          | (17.89)   |          |        | (22.46)   |           |
| avg_hhsize       |        | 6.703*    | 9.651*    |          | 0         | 0        |        | 6.723*    | 9.646*    |
|                  |        | (2.06)    | (2.57)    |          | (.)       | (.)      |        | (2.03)    | (2.56)    |
| gender_ratio     |        | 0.0691    | 0.496***  |          | 0         | 0        |        | 0.0661    | 0.495***  |
|                  |        | (0.75)    | (4.78)    |          | (.)       | (.)      |        | (0.70)    | (4.75)    |
| literacy_rate    |        | 0.821**   | -0.150    |          | 0         | 0        |        | 0.828**   | -0.152    |
|                  |        | (2.73)    | (-0.44)   |          | (.)       | (.)      |        | (2.71)    | (-0.45)   |
| urban_rate       |        | 0.920***  | 0.678***  |          | 0         | 0        |        | 0.919***  | 0.677***  |
|                  |        | (7.38)    | (4.74)    |          | (.)       | (.)      |        | (7.25)    | (4.72)    |
| dependency_ratio |        | 0.739     | -5.365*** |          | 0         | 0        |        | 0.729     | -5.374*** |
|                  |        | (0.54)    | (-3.51)   |          | (.)       | (.)      |        | (0.52)    | (-3.50)   |
| popul_density    |        | 0.00316   | -0.000863 |          | 0         | 0        |        | 0.00317   | -0.000870 |
|                  |        | (1.24)    | (-0.30)   |          | (.)       | (.)      |        | (1.22)    | (-0.30)   |
| covid_year       |        | -10.24*** | 14.66***  |          | -15.76*** | 9.735**  |        | -10.40*** | 14.63***  |
|                  |        | (-3.31)   | (4.51)    |          | (-4.49)   | (2.75)   |        | (-3.37)   | (4.51)    |
| _cons            | 20.47* | -200.6    | -371.4**  | 64.15*** | 33.74*    | 61.05*** | 22.29* | -197.1    | -370.2**  |
|                  | (2.00) | (-1.61)   | (-2.61)   | (4.13)   | (2.10)    | (3.34)   | (2.15) | (-1.55)   | (-2.59)   |
| N                | 1837   | 1423      | 1438      | 1837     | 1423      | 1438     | 1837   | 1423      | 1438      |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table A.5: Effect of social audits on delayed wage payment**

| Delayed wage             | Pooled OLS            |                      |                      | Fixed Effects        |                      |                      | Random Effects        |                      |                      |
|--------------------------|-----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|-----------------------|----------------------|----------------------|
|                          | (1)                   | (2)                  | (3)                  | (1)                  | (2)                  | (3)                  | (1)                   | (2)                  | (3)                  |
| pc_gp_audited_once       | -0.0773<br>(-0.44)    | -0.147<br>(-0.68)    | -0.128<br>(-0.60)    | -0.119<br>(-0.45)    | -0.190<br>(-0.56)    | -0.211<br>(-0.63)    | -0.0943<br>(-0.52)    | -0.164<br>(-0.74)    | -0.144<br>(-0.65)    |
| pc_gp_audited_twice      | -0.220<br>(-1.29)     | -0.183<br>(-0.85)    | -0.162<br>(-0.75)    | -0.239<br>(-0.92)    | -0.188<br>(-0.56)    | -0.211<br>(-0.64)    | -0.235<br>(-1.34)     | -0.193<br>(-0.87)    | -0.170<br>(-0.77)    |
| sa_expense_pergp         | 0.00273***<br>(3.74)  | 0.00210<br>(1.44)    | 0.00324***<br>(3.34) | 0.00414***<br>(4.20) | 0.00141<br>(0.79)    | 0.00312*<br>(2.38)   | 0.00290***<br>(3.92)  | 0.00190<br>(1.31)    | 0.00316**<br>(3.23)  |
| social_audited_amt_pergp | 0.00000248*<br>(2.51) | 0.00000116<br>(0.86) | 0.00000164<br>(1.29) | 0.00000126<br>(0.91) | 0.00000132<br>(0.73) | 0.00000233<br>(1.34) | 0.00000238*<br>(2.37) | 0.00000124<br>(0.91) | 0.00000177<br>(1.37) |
| works_audited_pergp      | 0.0571<br>(1.34)      | 0.122*<br>(2.22)     | 0.145**<br>(2.70)    | 0.143**<br>(2.62)    | 0.191**<br>(2.74)    | 0.199**<br>(2.90)    | 0.0694<br>(1.62)      | 0.132*<br>(2.40)     | 0.154**<br>(2.86)    |
| vsa_pergp                |                       | -1.621<br>(-0.34)    |                      |                      | -1.996<br>(-0.34)    |                      |                       | -1.552<br>(-0.33)    |                      |
| drp_pergp                |                       | 18.75*<br>(2.16)     |                      |                      | 31.14**<br>(2.94)    |                      |                       | 21.00*<br>(2.44)     |                      |
| total_jury_pergp         |                       | 13.95<br>(1.20)      |                      |                      | 15.03<br>(0.94)      |                      |                       | 14.04<br>(1.19)      |                      |
| sa_met_jcholders_pergp   |                       | 0.0163<br>(0.84)     |                      |                      | 0.0354<br>(1.36)     |                      |                       | 0.0176<br>(0.91)     |                      |

|                  |                 |                      |                      |                   |                     |                    |                 |                     |                     |
|------------------|-----------------|----------------------|----------------------|-------------------|---------------------|--------------------|-----------------|---------------------|---------------------|
| avg_hhsize       |                 | -0.277<br>(-0.04)    | 0.502<br>(0.07)      |                   | 0<br>(.)            | 0<br>(.)           |                 | -0.401<br>(-0.05)   | 0.420<br>(0.06)     |
| gender_ratio     |                 | 0.976***<br>(4.83)   | 0.969***<br>(5.00)   |                   | 0<br>(.)            | 0<br>(.)           |                 | 0.987***<br>(4.52)  | 0.974***<br>(4.63)  |
| literacy_rate    |                 | -0.690<br>(-1.07)    | -0.735<br>(-1.18)    |                   | 0<br>(.)            | 0<br>(.)           |                 | -0.688<br>(-0.99)   | -0.735<br>(-1.09)   |
| urban_rate       |                 | 0.627*<br>(2.32)     | 0.603*<br>(2.27)     |                   | 0<br>(.)            | 0<br>(.)           |                 | 0.617*<br>(2.10)    | 0.596*<br>(2.06)    |
| dependency_ratio |                 | 0.464<br>(0.15)      | -0.589<br>(-0.21)    |                   | 0<br>(.)            | 0<br>(.)           |                 | 0.555<br>(0.17)     | -0.534<br>(-0.17)   |
| popul_density    |                 | -0.00569<br>(-1.05)  | -0.00505<br>(-0.95)  |                   | 0<br>(.)            | 0<br>(.)           |                 | -0.00564<br>(-0.95) | -0.00501<br>(-0.86) |
| covid_year       |                 | -16.56*<br>(-2.50)   | -13.11*<br>(-2.20)   |                   | -21.32**<br>(-2.83) | -14.95*<br>(-2.30) |                 | -17.37**<br>(-2.68) | -13.62*<br>(-2.35)  |
| _cons            | 2.601<br>(0.15) | -959.5***<br>(-3.51) | -916.4***<br>(-3.43) | -5.086<br>(-0.20) | -7.682<br>(-0.23)   | 7.465<br>(0.23)    | 2.726<br>(0.16) | -972.5**<br>(-3.28) | -922.4**<br>(-3.18) |
| <i>N</i>         | 1716            | 1332                 | 1347                 | 1716              | 1332                | 1347               | 1716            | 1332                | 1347                |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table A.6: Effect of social audits on recovery done in SA**

| Recovery in SA           | Pooled OLS            |                       |                       | Fixed Effects       |                       |                        | Random Effects        |                       |                       |
|--------------------------|-----------------------|-----------------------|-----------------------|---------------------|-----------------------|------------------------|-----------------------|-----------------------|-----------------------|
|                          | (1)                   | (2)                   | (3)                   | (1)                 | (2)                   | (3)                    | (1)                   | (2)                   | (3)                   |
| pc_gp_audited_once       | -5.020<br>(-0.78)     | -6.566<br>(-1.03)     | -6.710<br>(-1.06)     | 0.133<br>(0.01)     | -4.248<br>(-0.39)     | -5.343<br>(-0.50)      | -5.020<br>(-0.78)     | -6.566<br>(-1.03)     | -6.710<br>(-1.06)     |
| pc_gp_audited_twice      | -6.495<br>(-1.04)     | -7.183<br>(-1.12)     | -6.217<br>(-0.97)     | -4.334<br>(-0.42)   | -6.663<br>(-0.62)     | -6.937<br>(-0.66)      | -6.495<br>(-1.04)     | -7.183<br>(-1.12)     | -6.217<br>(-0.97)     |
| sa_expense_pergp         | 0.0908***<br>(3.35)   | 0.0853*<br>(1.98)     | 0.0770**<br>(2.69)    | 0.0354<br>(0.89)    | 0.0499<br>(0.88)      | 0.0447<br>(1.08)       | 0.0908***<br>(3.35)   | 0.0853*<br>(1.98)     | 0.0770**<br>(2.69)    |
| social_audited_amt_pergp | -0.0000167<br>(-0.47) | -0.0000484<br>(-1.23) | -0.0000574<br>(-1.54) | 0.0000313<br>(0.58) | -0.0000111<br>(-0.20) | -0.00000710<br>(-0.13) | -0.0000167<br>(-0.47) | -0.0000484<br>(-1.23) | -0.0000574<br>(-1.54) |
| works_audited_pergp      | 2.662<br>(1.72)       | 1.949<br>(1.22)       | 2.751<br>(1.75)       | 0.656<br>(0.30)     | 1.007<br>(0.46)       | 1.317<br>(0.62)        | 2.662<br>(1.72)       | 1.949<br>(1.22)       | 2.751<br>(1.75)       |
| vsa_pergp                |                       | -290.8*<br>(-2.06)    |                       |                     | -225.0<br>(-1.22)     |                        |                       | -290.8*<br>(-2.06)    |                       |
| drp_pergp                |                       | 883.0***<br>(3.47)    |                       |                     | 714.7*<br>(2.13)      |                        |                       | 883.0***<br>(3.47)    |                       |
| total_jury_pergp         |                       | 367.0<br>(1.06)       |                       |                     | 321.9<br>(0.63)       |                        |                       | 367.0<br>(1.06)       |                       |
| sa_met_jcholders_pergp   |                       | -0.423                |                       |                     | 0.00290               |                        |                       | -0.423                |                       |

|                  |         |         |         |        |         |         |        |         |         |
|------------------|---------|---------|---------|--------|---------|---------|--------|---------|---------|
|                  |         | (-0.76) |         |        | (0.00)  |         |        | (-0.76) |         |
| avg_hhsize       | 70.80   | 100.7   |         |        | 0       | 0       |        | 70.80   | 100.7   |
|                  | (0.35)  | (0.50)  |         |        | (.)     | (.)     |        | (0.35)  | (0.50)  |
| gender_ratio     | 3.361   | 1.177   |         |        | 0       | 0       |        | 3.361   | 1.177   |
|                  | (0.58)  | (0.21)  |         |        | (.)     | (.)     |        | (0.58)  | (0.21)  |
| literacy_rate    | -12.20  | -12.52  |         |        | 0       | 0       |        | -12.20  | -12.52  |
|                  | (-0.65) | (-0.69) |         |        | (.)     | (.)     |        | (-0.65) | (-0.69) |
| urban_rate       | 9.909   | 11.24   |         |        | 0       | 0       |        | 9.909   | 11.24   |
|                  | (1.27)  | (1.46)  |         |        | (.)     | (.)     |        | (1.27)  | (1.46)  |
| dependency_ratio | 136.7   | 138.0   |         |        | 0       | 0       |        | 136.7   | 138.0   |
|                  | (1.59)  | (1.68)  |         |        | (.)     | (.)     |        | (1.59)  | (1.68)  |
| popul_density    | 0.0439  | 0.0512  |         |        | 0       | 0       |        | 0.0439  | 0.0512  |
|                  | (0.28)  | (0.33)  |         |        | (.)     | (.)     |        | (0.28)  | (0.33)  |
| covid_year       | -278.8  | -167.5  |         |        | -247.0  | -125.9  |        | -278.8  | -167.5  |
|                  | (-1.44) | (-0.96) |         |        | (-1.05) | (-0.62) |        | (-1.44) | (-0.96) |
| _cons            | 267.2   | -6944.2 | -4903.2 | 169.1  | 347.3   | 571.9   | 267.2  | -6944.2 | -4903.2 |
|                  | (0.43)  | (-0.89) | (-0.64) | (0.16) | (0.32)  | (0.55)  | (0.43) | (-0.89) | (-0.64) |
| <i>N</i>         | 1837    | 1423    | 1438    | 1837   | 1423    | 1438    | 1837   | 1423    | 1438    |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table A.7: Effect of social audits on penalty imposed on transgressors**

| Penalty imposed during<br>SA | Pooled OLS              |                         |                         | Fixed Effects          |                         |                         | Random Effects          |                         |                         |
|------------------------------|-------------------------|-------------------------|-------------------------|------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
|                              | (1)                     | (2)                     | (3)                     | (1)                    | (2)                     | (3)                     | (1)                     | (2)                     | (3)                     |
| pc_gp_audited_once           | 0.188<br>(0.71)         | 0.165<br>(0.72)         | 0.163<br>(0.72)         | 0.0195<br>(0.05)       | 0.0284<br>(0.07)        | 0.0157<br>(0.04)        | 0.188<br>(0.71)         | 0.165<br>(0.72)         | 0.163<br>(0.72)         |
| pc_gp_audited_twice          | 0.116<br>(0.45)         | 0.116<br>(0.51)         | 0.114<br>(0.50)         | -0.0152<br>(-0.04)     | 0.0716<br>(0.19)        | 0.0540<br>(0.14)        | 0.116<br>(0.45)         | 0.116<br>(0.51)         | 0.114<br>(0.50)         |
| sa_expense_pergp             | 0.0000246<br>(0.02)     | 0.000479<br>(0.31)      | 0.000717<br>(0.70)      | -0.000810<br>(-0.50)   | -0.00150<br>(-0.74)     | -0.000816<br>(-0.55)    | 0.0000246<br>(0.02)     | 0.000479<br>(0.31)      | 0.000717<br>(0.70)      |
| social_audited_amt_pergp     | -0.000000673<br>(-0.46) | -0.000000926<br>(-0.66) | -0.000000783<br>(-0.59) | -0.00000103<br>(-0.46) | -0.000000429<br>(-0.21) | -0.000000145<br>(-0.08) | -0.000000673<br>(-0.46) | -0.000000926<br>(-0.66) | -0.000000783<br>(-0.59) |
| works_audited_pergp          | -0.0886<br>(-1.40)      | -0.0643<br>(-1.12)      | -0.0613<br>(-1.09)      | -0.0705<br>(-0.79)     | -0.0457<br>(-0.59)      | -0.0431<br>(-0.56)      | -0.0886<br>(-1.40)      | -0.0643<br>(-1.12)      | -0.0613<br>(-1.09)      |
| vsa_pergp                    |                         | 0.426<br>(0.08)         |                         |                        | 1.014<br>(0.15)         |                         |                         | 0.426<br>(0.08)         |                         |
| drp_pergp                    |                         | -1.490<br>(-0.16)       |                         |                        | 1.042<br>(0.09)         |                         |                         | -1.490<br>(-0.16)       |                         |
| total_jury_pergp             |                         | 4.746<br>(0.38)         |                         |                        | 8.179<br>(0.44)         |                         |                         | 4.746<br>(0.38)         |                         |
| sa_met_jcholders_pergp       |                         | 0.00795                 |                         |                        | 0.0195                  |                         |                         | 0.00795                 |                         |

|                  |                 |                     |                     |                 |                   |                   |                 |                     |                     |
|------------------|-----------------|---------------------|---------------------|-----------------|-------------------|-------------------|-----------------|---------------------|---------------------|
|                  |                 | (0.40)              |                     |                 | (0.67)            |                   |                 | (0.40)              |                     |
| avg_hhsize       |                 | -6.230<br>(-0.85)   | -5.899<br>(-0.82)   |                 | 0<br>(.)          | 0<br>(.)          |                 | -6.230<br>(-0.85)   | -5.899<br>(-0.82)   |
| gender_ratio     |                 | 0.140<br>(0.68)     | 0.166<br>(0.83)     |                 | 0<br>(.)          | 0<br>(.)          |                 | 0.140<br>(0.68)     | 0.166<br>(0.83)     |
| literacy_rate    |                 | -0.939<br>(-1.39)   | -0.934<br>(-1.44)   |                 | 0<br>(.)          | 0<br>(.)          |                 | -0.939<br>(-1.39)   | -0.934<br>(-1.44)   |
| urban_rate       |                 | 0.666*<br>(2.38)    | 0.633*<br>(2.31)    |                 | 0<br>(.)          | 0<br>(.)          |                 | 0.666*<br>(2.38)    | 0.633*<br>(2.31)    |
| dependency_ratio |                 | 0.218<br>(0.07)     | -0.288<br>(-0.10)   |                 | 0<br>(.)          | 0<br>(.)          |                 | 0.218<br>(0.07)     | -0.288<br>(-0.10)   |
| popul_density    |                 | -0.00577<br>(-1.01) | -0.00524<br>(-0.94) |                 | 0<br>(.)          | 0<br>(.)          |                 | -0.00577<br>(-1.01) | -0.00524<br>(-0.94) |
| covid_year       |                 | 0.755<br>(0.11)     | 1.347<br>(0.22)     |                 | -8.030<br>(-0.95) | -6.474<br>(-0.89) |                 | 0.755<br>(0.11)     | 1.347<br>(0.22)     |
| __cons           | 2.273<br>(0.09) | -86.89<br>(-0.31)   | -96.69<br>(-0.35)   | 21.59<br>(0.51) | 12.96<br>(0.33)   | 17.46<br>(0.47)   | 2.273<br>(0.09) | -86.89<br>(-0.31)   | -96.69<br>(-0.35)   |
| <i>N</i>         | 1837            | 1423                | 1438                | 1837            | 1423              | 1438              | 1837            | 1423                | 1438                |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table A.8: Effect of social audits on records not submitted**

| Records not submitted    | Pooled OLS                    |                               |                               | Fixed Effects                 |                               |                               | Random Effects                |                               |                               |
|--------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                          | (1)                           | (2)                           | (3)                           | (1)                           | (2)                           | (3)                           | (1)                           | (2)                           | (3)                           |
| pc_gp_audited_once       | 0.0280<br>(0.72)              | 0.0471<br>(1.07)              | 0.0500<br>(1.14)              | 0.0675<br>(1.08)              | 0.0682<br>(0.91)              | 0.0677<br>(0.91)              | 0.0280<br>(0.72)              | 0.0471<br>(1.07)              | 0.0500<br>(1.14)              |
| pc_gp_audited_twice      | -0.0500<br>(-1.31)            | 0.00762<br>(0.17)             | 0.0204<br>(0.46)              | -0.0327<br>(-0.54)            | 0.0172<br>(0.23)              | 0.0264<br>(0.36)              | -0.0500<br>(-1.31)            | 0.00762<br>(0.17)             | 0.0204<br>(0.46)              |
| sa_expense_pergp         | -0.000333*<br>(-2.03)         | -0.000968**<br>(-3.26)        | -0.000403*<br>(-2.04)         | -0.000654**<br>(-2.79)        | -0.00127**<br>(-3.29)         | -0.00108***<br>(-3.76)        | -0.000333*<br>(-2.03)         | -0.000968**<br>(-3.26)        | -0.000403*<br>(-2.04)         |
| social_audited_amt_pergp | -<br>0.00000156***<br>(-7.18) | -<br>0.00000141***<br>(-5.21) | -<br>0.00000170***<br>(-6.61) | -<br>0.00000247***<br>(-7.71) | -<br>0.00000176***<br>(-4.60) | -<br>0.00000198***<br>(-5.34) | -<br>0.00000156***<br>(-7.18) | -<br>0.00000141***<br>(-5.21) | -<br>0.00000170***<br>(-6.61) |
| works_audited_pergp      | 0.0275**<br>(2.93)            | 0.0325**<br>(2.94)            | 0.0347**<br>(3.19)            | 0.0336**<br>(2.61)            | 0.0395**<br>(2.65)            | 0.0347*<br>(2.35)             | 0.0275**<br>(2.93)            | 0.0325**<br>(2.94)            | 0.0347**<br>(3.19)            |
| vsa_pergp                |                               | 1.947*<br>(2.00)              |                               |                               | 0.606<br>(0.48)               |                               |                               | 1.947*<br>(2.00)              |                               |
| drp_pergp                |                               | 6.398***<br>(3.65)            |                               |                               | 6.339**<br>(2.75)             |                               |                               | 6.398***<br>(3.65)            |                               |
| total_jury_pergp         |                               | 3.707<br>(1.55)               |                               |                               | 1.530<br>(0.44)               |                               |                               | 3.707<br>(1.55)               |                               |
| sa_met_jcholders_pergp   |                               | -0.0124**                     |                               |                               | -0.0182**                     |                               |                               | -0.0124**                     |                               |

|                  |         |           |           |        |          |           |         |           |           |
|------------------|---------|-----------|-----------|--------|----------|-----------|---------|-----------|-----------|
|                  |         | (-3.20)   |           |        | (-3.26)  |           |         | (-3.20)   |           |
| avg_hhsize       |         | 3.237*    | 2.980*    |        | 0        | 0         |         | 3.237*    | 2.980*    |
|                  |         | (2.31)    | (2.13)    |        | (.)      | (.)       |         | (2.31)    | (2.13)    |
| gender_ratio     |         | 0.0762    | 0.0357    |        | 0        | 0         |         | 0.0762    | 0.0357    |
|                  |         | (1.91)    | (0.92)    |        | (.)      | (.)       |         | (1.91)    | (0.92)    |
| literacy_rate    |         | -0.495*** | -0.436*** |        | 0        | 0         |         | -0.495*** | -0.436*** |
|                  |         | (-3.83)   | (-3.46)   |        | (.)      | (.)       |         | (-3.83)   | (-3.46)   |
| urban_rate       |         | -0.207*** | -0.203*** |        | 0        | 0         |         | -0.207*** | -0.203*** |
|                  |         | (-3.86)   | (-3.81)   |        | (.)      | (.)       |         | (-3.86)   | (-3.81)   |
| dependency_ratio |         | -1.666**  | -1.509**  |        | 0        | 0         |         | -1.666**  | -1.509**  |
|                  |         | (-2.82)   | (-2.65)   |        | (.)      | (.)       |         | (-2.82)   | (-2.65)   |
| popul_density    |         | -0.00204  | -0.00149  |        | 0        | 0         |         | -0.00204  | -0.00149  |
|                  |         | (-1.86)   | (-1.38)   |        | (.)      | (.)       |         | (-1.86)   | (-1.38)   |
| covid_year       |         | -4.558*** | -6.156*** |        | -5.019** | -6.605*** |         | -4.558*** | -6.156*** |
|                  |         | (-3.42)   | (-5.09)   |        | (-3.10)  | (-4.69)   |         | (-3.42)   | (-5.09)   |
| __cons           | 11.22** | 9.330     | 42.07     | 10.93  | 13.44    | 12.76     | 11.22** | 9.330     | 42.07     |
|                  | (2.99)  | (0.17)    | (0.80)    | (1.79) | (1.81)   | (1.75)    | (2.99)  | (0.17)    | (0.80)    |
| N                | 1837    | 1423      | 1438      | 1837   | 1423     | 1438      | 1837    | 1423      | 1438      |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table A.9: Effect of social audits on total issues reported**

| Total issues reported        | Pooled OLS            |                       |                       | Fixed Effects         |                      |                       | Random Effects        |                       |                       |
|------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                              | (1)                   | (2)                   | (3)                   | (1)                   | (2)                  | (3)                   | (1)                   | (2)                   | (3)                   |
| pc_gp_audited_once           | 0.00886<br>(0.08)     | 0.0487<br>(0.31)      | 0.0529<br>(0.34)      | 0.00736<br>(0.04)     | 0.120<br>(0.52)      | 0.112<br>(0.49)       | 0.0147<br>(0.12)      | 0.0718<br>(0.42)      | 0.0686<br>(0.41)      |
| pc_gp_audited_twice          | -0.0241<br>(-0.21)    | 0.0274<br>(0.17)      | 0.0174<br>(0.11)      | -0.0371<br>(-0.23)    | 0.0615<br>(0.27)     | 0.0467<br>(0.21)      | -0.0254<br>(-0.21)    | 0.0353<br>(0.21)      | 0.0199<br>(0.12)      |
| sa_expense_pergp             | 0.000850<br>(1.72)    | 0.000647<br>(0.61)    | 0.000412<br>(0.59)    | 0.00137*<br>(2.19)    | 0.00203<br>(1.68)    | 0.00151<br>(1.72)     | 0.00104*<br>(2.04)    | 0.00117<br>(1.13)     | 0.000793<br>(1.11)    |
| social_audited_amt_perg<br>p | 0.000000362<br>(0.56) | 0.000000219<br>(0.23) | 0.000000575<br>(0.63) | 0.000000501<br>(0.59) | -7.68e-08<br>(-0.06) | 0.000000190<br>(0.17) | 0.000000391<br>(0.57) | 0.000000134<br>(0.14) | 0.000000444<br>(0.47) |
| works_audited_pergp          | 0.0287<br>(1.02)      | 0.0411<br>(1.04)      | 0.0416<br>(1.08)      | 0.0128<br>(0.38)      | 0.0143<br>(0.31)     | 0.0195<br>(0.43)      | 0.0219<br>(0.77)      | 0.0300<br>(0.77)      | 0.0313<br>(0.81)      |
| vsa_pergp                    |                       | -1.870<br>(-0.54)     |                       |                       | -3.789<br>(-0.97)    |                       |                       | -2.716<br>(-0.80)     |                       |
| drp_pergp                    |                       | -2.946<br>(-0.47)     |                       |                       | -6.256<br>(-0.87)    |                       |                       | -4.563<br>(-0.74)     |                       |
| total_jury_pergp             |                       | -7.769<br>(-0.91)     |                       |                       | -2.220<br>(-0.20)    |                       |                       | -6.567<br>(-0.75)     |                       |
| sa_met_jcholders_pergp       |                       | 0.0176                |                       |                       | 0.0251               |                       |                       | 0.0200                |                       |

|                  |                 |                    |                   |                 |                   |                   |                 |                      |                   |
|------------------|-----------------|--------------------|-------------------|-----------------|-------------------|-------------------|-----------------|----------------------|-------------------|
|                  |                 | (1.28)             |                   |                 | (1.45)            |                   |                 | (1.41)               |                   |
| avg_hhsize       |                 | 2.813<br>(0.56)    | 2.986<br>(0.61)   |                 | 0<br>(.)          | 0<br>(.)          |                 | 3.141<br>(0.47)      | 3.463<br>(0.51)   |
| gender_ratio     |                 | -0.0142<br>(-0.10) | 0.0230<br>(0.17)  |                 | 0<br>(.)          | 0<br>(.)          |                 | -0.000881<br>(-0.00) | 0.0536<br>(0.29)  |
| literacy_rate    |                 | 0.992*<br>(2.15)   | 0.830<br>(1.87)   |                 | 0<br>(.)          | 0<br>(.)          |                 | 1.124<br>(1.85)      | 0.967<br>(1.63)   |
| urban_rate       |                 | 0.232<br>(1.21)    | 0.220<br>(1.17)   |                 | 0<br>(.)          | 0<br>(.)          |                 | 0.278<br>(1.08)      | 0.262<br>(1.03)   |
| dependency_ratio |                 | 5.877**<br>(2.79)  | 5.468**<br>(2.73) |                 | 0<br>(.)          | 0<br>(.)          |                 | 6.926*<br>(2.48)     | 6.482*<br>(2.38)  |
| popul_density    |                 | 0.00830*<br>(2.13) | 0.00731<br>(1.92) |                 | 0<br>(.)          | 0<br>(.)          |                 | 0.0104*<br>(1.97)    | 0.00935<br>(1.79) |
| covid_year       |                 | -3.808<br>(-0.80)  | -1.099<br>(-0.26) |                 | -2.524<br>(-0.50) | 1.174<br>(0.27)   |                 | -3.445<br>(-0.77)    | -0.307<br>(-0.08) |
| _cons            | 2.929<br>(0.26) | -247.2<br>(-1.29)  | -262.0<br>(-1.41) | 1.451<br>(0.09) | -9.850<br>(-0.43) | -9.141<br>(-0.41) | 2.170<br>(0.18) | -306.9<br>(-1.21)    | -338.2<br>(-1.34) |
| <i>N</i>         | 1837            | 1423               | 1438              | 1837            | 1423              | 1438              | 1837            | 1423                 | 1438              |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

**Table A.10: Effect of social audits on number of issues resolved**

| Issued resolved          | Pooled OLS          |                       |                       | Fixed Effects         |                         |                         | Random Effects      |                       |                       |
|--------------------------|---------------------|-----------------------|-----------------------|-----------------------|-------------------------|-------------------------|---------------------|-----------------------|-----------------------|
|                          | (1)                 | (2)                   | (3)                   | (1)                   | (2)                     | (3)                     | (1)                 | (2)                   | (3)                   |
| pc_gp_audited_once       | 0.0542<br>(1.06)    | 0.0242<br>(0.68)      | 0.0254<br>(0.71)      | 0.0367<br>(0.44)      | 0.0194<br>(0.33)        | 0.0248<br>(0.43)        | 0.0542<br>(1.06)    | 0.0243<br>(0.67)      | 0.0256<br>(0.71)      |
| pc_gp_audited_twice      | 0.0260<br>(0.52)    | -0.000421<br>(-0.01)  | 0.00289<br>(0.08)     | 0.00311<br>(0.04)     | 0.00563<br>(0.10)       | 0.0142<br>(0.25)        | 0.0260<br>(0.52)    | 0.000322<br>(0.01)    | 0.00389<br>(0.11)     |
| sa_expense_pergp         | 0.000220<br>(1.02)  | -0.000234<br>(-0.97)  | 0.000115<br>(0.72)    | 0.000109<br>(0.35)    | -0.000345<br>(-1.13)    | 0.000106<br>(0.47)      | 0.000220<br>(1.02)  | -0.000242<br>(-1.00)  | 0.000114<br>(0.71)    |
| social_audited_amt_pergp | 2.52e-09<br>(0.01)  | 0.000000386<br>(1.74) | 0.000000252<br>(1.20) | 0.000000708<br>(1.66) | 0.000000966**<br>(3.20) | 0.000000872**<br>(2.99) | 2.52e-09<br>(0.01)  | 0.000000415<br>(1.87) | 0.000000281<br>(1.34) |
| works_audited_pergp      | -0.00727<br>(-0.59) | -0.0131<br>(-1.45)    | -0.0137<br>(-1.56)    | -0.0254<br>(-1.49)    | -0.0298*<br>(-2.54)     | -0.0300**<br>(-2.58)    | -0.00727<br>(-0.59) | -0.0139<br>(-1.54)    | -0.0145<br>(-1.64)    |
| vsa_pergp                |                     | 2.279**<br>(2.87)     |                       |                       | 3.157**<br>(3.17)       |                         |                     | 2.327**<br>(2.93)     |                       |
| drp_pergp                |                     | 0.256<br>(0.18)       |                       |                       | 1.604<br>(0.88)         |                         |                     | 0.318<br>(0.22)       |                       |
| total_jury_pergp         |                     | 2.672<br>(1.37)       |                       |                       | 3.815<br>(1.38)         |                         |                     | 2.735<br>(1.40)       |                       |
| sa_met_jcholders_pergp   |                     | -0.00596              |                       |                       | -0.00904*               |                         |                     | -0.00612              |                       |

|                  |                   |                    |                    |                 |                 |                   |                   |                    |                    |
|------------------|-------------------|--------------------|--------------------|-----------------|-----------------|-------------------|-------------------|--------------------|--------------------|
|                  |                   | (-1.90)            |                    |                 | (-2.06)         |                   |                   | (-1.94)            |                    |
| avg_hhsize       |                   | -0.0454<br>(-0.04) | -0.323<br>(-0.28)  |                 | 0<br>(.)        | 0<br>(.)          |                   | -0.0204<br>(-0.02) | -0.304<br>(-0.26)  |
| gender_ratio     |                   | 0.0357<br>(1.10)   | 0.0189<br>(0.60)   |                 | 0<br>(.)        | 0<br>(.)          |                   | 0.0350<br>(1.05)   | 0.0176<br>(0.55)   |
| literacy_rate    |                   | -0.114<br>(-1.08)  | -0.0681<br>(-0.67) |                 | 0<br>(.)        | 0<br>(.)          |                   | -0.113<br>(-1.05)  | -0.0654<br>(-0.62) |
| urban_rate       |                   | 0.0648<br>(1.48)   | 0.0584<br>(1.35)   |                 | 0<br>(.)        | 0<br>(.)          |                   | 0.0650<br>(1.45)   | 0.0588<br>(1.32)   |
| dependency_ratio |                   | -0.737<br>(-1.53)  | -0.716<br>(-1.55)  |                 | 0<br>(.)        | 0<br>(.)          |                   | -0.756<br>(-1.53)  | -0.731<br>(-1.54)  |
| popul_density    |                   | 0.00146<br>(1.64)  | 0.00208*<br>(2.38) |                 | 0<br>(.)        | 0<br>(.)          |                   | 0.00145<br>(1.58)  | 0.00208*<br>(2.31) |
| covid_year       |                   | 1.089<br>(1.00)    | -0.423<br>(-0.43)  |                 | 0.921<br>(0.72) | -1.301<br>(-1.18) |                   | 1.070<br>(0.99)    | -0.480<br>(-0.49)  |
| __cons           | -1.681<br>(-0.34) | -7.055<br>(-0.16)  | 7.381<br>(0.17)    | 0.591<br>(0.07) | 1.184<br>(0.20) | 1.391<br>(0.24)   | -1.681<br>(-0.34) | -5.965<br>(-0.13)  | 8.815<br>(0.20)    |
| <i>N</i>         | 1837              | 1423               | 1438               | 1837            | 1423            | 1438              | 1837              | 1423               | 1438               |

*t* statistics in parentheses

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$